WESTCLIFF UNIVERSITY
Educate. Inspire. Empower.

DATA 300 Data Driven Decision Making

Units: 3
Modality: async / synch / on campus
Length Of Class: 8 Weeks
Prerequisites: None

Student Note: Some links may function best when the syllabus is downloaded and viewed on your device rather than in a web browser.

Course Description

This course covers a variety of data driven decision making topics for the purpose of making better strategic business decisions. Topics include BI tools and data analytics, with emphasis on predictive analytics focused on information and knowledge pertaining to customers, competitors, internal operations, and external variables.

Course Learning Outcomes

Learning outcomes are statements that describe significant and essential scholarship that students achieve and can reliably demonstrate by the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course; the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Outcomes (CLOs)	Aligned Program Learning Outcomes (PLOs)
CLO 1 Describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges.	PLO: 4, 5
CLO 2 Design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions.	PLO: 4, 5
CLO 3 Apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data.	PLO: 1, 2, 4, 5
CLO 4 Develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes.	PLO: 1, 2, 4
CLO 5 Evaluate the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts.	PLO: 3, 6

Course Learning Materials

Primary Resources
Mary Ellen Gordon (2023). Business Analytics: Combining data, analysis and judgement to inform decisions . SAGE Publications Ltd. ISBN: 9781529615999, 9781529796117.

https://perlego.com/book/4336901/business-analytics-combining-data-analysis-and-judgement-to-inform-decisions-pdf/?utm_medium=share&utm_source=perlego&utm_campaign=share-book

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Weekly Course Learning Overview & Summary of Graded Work

The *Weekly Course Learning Overview & Summary of Graded Work Table* provides a structured breakdown of the course by weekly units, outlining assigned learning materials, graded activities, and the course learning outcomes addressed each week. This resource is designed to help students clearly see how each component of the course contributes to their overall learning and progression. By reviewing this table regularly, students can effectively plan their workload, stay on track with deadlines, and ensure meaningful engagement with course content. It serves as a roadmap for the term, promoting consistency, organization, and purposeful learning throughout the learning process.

Each graded assignment in this course is intentionally designed to align with the established course learning outcomes, ensuring that students develop and demonstrate the knowledge and skills required for successful dissertation completion. The weight assigned to each graded item reflects its relative importance and impact on the final course grade. Reviewing the table provides clarity on how performance is evaluated and supports effective planning and time management. This structure helps students understand the purpose behind each task and how their work contributes to overall academic progress.

Grading criteria description and rubrics for the course are in the Appendix at the end of the syllabus.

Unit	Graded Work	Weight	CLOs
Week 1	Activity 1: Self-Assessment: JupyterLab Lab and Tableau Setup Readiness Check Applied Lab 1: Data Exploration	4% 4%	1, 2, 3, 4 1, 2, 3
Week 2	Applied Lab 2: Dashboard Creation Assignment 1: Data Visualization and Descriptive Analytics Report	4 % 8%	1, 2, 3 1, 2, 3
Week 3	Applied Lab 3: Text Mining Application SM Activity 1	4 % 3 %	1, 2, 3 1, 2, 3
Week 4	Applied Lab 4: Forecasting Midterm Exam	4 % 15 %	1, 2, 5 1, 2, 3, 4
Week 5	Applied Lab 5: Predictive Linear Regression Modeling Assignment 2: Predictive Data Mining Report	4 % 7 %	4, 5 1, 2, 3, 4
Week 6	Applied Lab 6: Advanced Spreadsheet Simulation	4 %	4, 5
Week 7	Applied Lab 7: BI Case Study Implementation Assignment 3: Spreadsheet Simulation and Decision Model	4 % 7 %	4, 5 3, 4, 5

Week 8	Assignment 4: Final Comprehensive Assignment (Capstone) SM Activity 2 Reflection	25 % 3 %	4, 5 1, 2, 3, 4, 5
		100%	

APPENDIX

The Appendix provides additional resources and key policies to support student success and ensure compliance with institutional and accreditation standards. It includes Westcliff University and course-specific policies, as well as a Detailed Course Outline, which presents a comprehensive breakdown of the course by week. Additional sections in the Appendix include:

- Policies
- Detailed Course Outline
- Course Rubrics
- Westcliff University Library & Digital Resources Center
- Accessibility and ADA Compliance
- Copyright Notice of Course Materials
- Course Learning Materials Reference List

Policies

Please review the Westcliff University Policies that apply to all students and courses. The Program Policies document speaks to specific policies relevant to the college and program.

Collaborative learning is a fundamental component of this course, and active participation in Collaborations will enrich your understanding and foster a vibrant learning community. Throughout the 8-weeks, you will have the opportunity to engage in both synchronous and asynchronous collaborations.

Course Collaborations Overview

This course may be delivered in either synchronous or asynchronous format. Collaboration expectations vary by modality as outlined below.

Synchronous Modality
In courses delivered synchronously, students participate in weekly live virtual class sessions. Students are expected to review assigned materials in advance and engage actively and professionally during class discussions and activities.

Asynchronous Modality
In courses delivered asynchronously, students participate in eight weekly online collaborations designed to extend course learning and deepen application of key concepts.

Requirements:

- Initial post due Thursday by 11:59 p.m. Pacific Time (250 to 300 words), demonstrating critical thinking and substantive engagement with the prompt.
- At least two peer responses due Sunday by 11:59 p.m. Pacific Time (100 to 150 words each), advancing the discussion in a professional and meaningful manner.
- All outside sources must be cited using APA 7th edition format with full references provided.
- Grading is based on the Collaboration Grading Rubric (see syllabus appendix or PDF).

- Active participation in the designated collaboration format is essential to a rigorous and engaging graduate learning experience.

Course Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment and reach out to your instructor for additional information as needed.

All assignments are to be submitted by ***Sunday at 11:59 p.m. Pacific Time.***

All written work must adhere to *APA 7th edition* academic formatting requirements including core components such as the cover page, page numbers, headings, citations, 1" margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents.

Student Engagement and Participation

Students are expected to actively engage in the course each week in accordance with the course modality (online asynchronous, online synchronous, or hybrid/on-campus) and the expectations outlined in the course policies.

Students receive a weekly grade based on their demonstrated participation in required course activities. Depending on the course modality, participation may include, but is not limited to:

- Contribute meaningfully to discussions or collaborative activities
- Ask and respond to questions
- Provide thoughtful comments or peer feedback
- Participate in group or team-based learning activities
- Engage with instructor-led learning experiences as required

For courses with required live sessions, students are expected to attend or view recordings as specified in the course policies.

Detailed Course Outline

The Detailed Course Outline offers a comprehensive, week-by-week overview of all course activities, including instructional guidance, required learning materials, supplemental resources, and grading criteria. This outline is designed to support students in understanding what is expected each week and how each assignment contributes to the demonstration of course learning outcomes (CLOs). Students are responsible for completing all listed assignments by the indicated deadlines to ensure steady progress and mastery of key competencies. For more specific expectations related to each assignment, students should refer to the Detailed Description of Each Grading Criteria section in the syllabus. This outline serves as a vital planning and reference tool to support academic success throughout the term.

All the grading rubrics for the course are provided in the syllabus in the ***Course Rubrics*** Section. (note this link may only work in the Downloaded PDF)

Week 1

Learning Materials:

Read

Gordon, M. E. (2023). Business analytics. SAGE Publications Ltd.

- Chapter 1: How Data and Analytics Can Enhance Human Decision-Making in Organizations
- Chapter 4: Analysis Fundamentals

Data-Driven Decision Making (2025): Latest Stats, Trends & Benchmarks, https://www.hydrogenbi.com/data-driven-decision-making-2025-stats

Qlik. (n.d.). *What is descriptive analytics?* https://www.qlik.com/us/reporting-analytics/descriptive-analytics

Analytics Vidhya. (2015). *Ultimate guide for data exploration in Python using NumPy, Matplotlib and Pandas*. https://www.analyticsvidhya.com/blog/2015/04/comprehensive-guide-data-exploration-sas-using-python-numpy-scipy-matplotlib-pandas/

Software / Platforms

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Watch

Google Data Analytics Certificate. (n.d.). *Making data-driven decisions in business* [Video]. YouTube. https://www.youtube.com/watch?v=EevYFzAqJ2M

What is data-driven decision making? 6 key steps, importance, tools & techniques. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=-4iIC5xsiGM

Data-driven decision making: How to go from data to decisions! (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=EV3T1RYWC9g

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

What are descriptive analytics? (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=DIFqQy10aCs

Descriptive, predictive, & prescriptive analytics. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=lxCaYg1G9fE

Activity 1 Self-Assessment: SAS Environment Setup, JupyterLab, and Readiness Check CLO 1, 2, 3, 4

Overview & Objectives

This self-check diagnostic lab allows you to evaluate your baseline understanding of analytics terminology and computational logic while ensuring your primary software tools are correctly configured for the semester.

In this lab, you will use JupyterLab (via the web-based Try Jupyter environment) to run a Python script, load the built-in Palmer Penguins dataset, perform basic Exploratory Data Analysis (EDA), and export your completed work as a PDF.

By the end of this lab, you will be able to:

- Access and navigate the JupyterLab web notebook environment.
- Install and import core data science libraries (pandas, seaborn, matplotlib).
- Execute Python code cells sequentially to inspect summary statistics and dataset previews.
- Interpret bivariate scatter plots and categorical boxplots.
- Export executed notebook outputs directly to PDF format.

Accessing JupyterLab

Part 1: Accessing JupyterLab
JupyterLab is an open-source, web-based interactive development environment for notebooks, code, and data. The Try Jupyter environment runs Python directly in your browser using WebAssembly

(Pyodide), requiring no installation.

- Open your web browser and navigate to: https://jupyter.org/try-jupyter/lab/
- Create a new notebook by clicking the **Python (Pyodide)** or **Python 3** icon under the **Notebook** section in the Launcher tab.
- Rename your notebook file:
  - Right-click the notebook tab or file name in the left panel -> Select **Rename Notebook...**
  - Name format: YourLastName_Data300_Self_Check.ipynb

Part 2: Executing Python Code for EDA
You do not need to write original code for this diagnostic setup. Your objective is to run the provided script block, which loads a dataset of penguin observations, outputs descriptive statistics, and generates two visual plots.

Note for Pyodide Users: Because the Pyodide kernel runs Python inside your browser via WebAssembly, you must install seaborn into the browser session using %pip install seaborn before importing it.

In a new **Code** cell, paste and run the following code block (press Shift + Enter or click the Play button in the notebook toolbar):

```
# 0. Install required library in Pyodide browser environment
%pip install seaborn

# 1. Import core analytics libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from pyodide.http import open_url

# 2. Load the 'penguins' dataset
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/penguins.csv"
df = pd.read_csv(open_url(url))

# 3. Basic Analytics: Display first 5 rows and summary statistics
print("--- First 5 rows of the dataset ---")
display(df.head())

print("\n--- Summary Statistics ---")
display(df.describe())

# 4. EDA Plot 1: Scatter plot of Flipper Length vs Body Mass
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=df,
    x='flipper_length_mm',
    y='body_mass_g',
```

```
    hue='species',
    palette='deep'
)
plt.title('Penguin Body Mass vs. Flipper Length by Species')
plt.xlabel('Flipper Length (mm)')
plt.ylabel('Body Mass (g)')
plt.show()

# 5. EDA Plot 2: Boxplot of Body Mass by Island and Sex
plt.figure(figsize=(8, 5))
sns.boxplot(
    data=df,
    x='island',
    y='body_mass_g',
    hue='sex',
    palette='pastel'
)
plt.title('Body Mass Distribution by Island and Sex')
plt.show()
```

Part 3: Diagnostic Analytical Questions
Based on the statistical output and data visualizations generated in your notebook, answer the following questions in a Markdown cell at the bottom of your notebook (or in your submission document as required by your instructor):

- **Descriptive Statistics:**
  - Inspect the **Summary Statistics** table output (df.describe()). What is the mean (average) body mass in grams (body_mass_g) for the penguins in this dataset?
- **Bivariate Correlation Trend:**
  - Examine **EDA Plot 1** (Scatter plot). As flipper length (flipper_length_mm) increases, what general pattern or trajectory occurs with body mass (body_mass_g)?
- **Categorical Group Comparison:**
  - Based on **EDA Plot 1**, which of the three penguin species (*Adelie*, *Chinstrap*, or *Gentoo*) displays the highest overall body mass cluster?

Part 4: Saving and Exporting Your Work
Step 1: Save and Export to PDF

1. Confirm that all code cells have executed sequentially and that all output tables and both plots are fully visible.
2. In the top JupyterLab menu bar, click File -> Print (or Save and Export Notebook As... -> HTML).
3. In your browser's print preview dialog box:

- Change the Destination to Save as PDF.
- Enable Background graphics to ensure visual colors on plots remain intact.
- Click Save and name your file Data300_Self_Check_Results_LastName.pdf.

Tableau Setup & Exploratory Visualization
Tableau is an industry-standard Business Intelligence (BI) tool. As a student, you get a free 1-year license to Tableau Desktop.
Step 1: Claiming Your Free Student License

1. Go to https://www.tableau.com/academic/students.

2. Click the Get Tableau for Free button.
3. Fill out the form using your university email address. You may be asked to verify your student status (usually via an automated system or by uploading a student ID).
4. Once verified, you will receive a product key via email. Download Tableau Desktop, install it, and activate it using the key provided.

Step 2: Connect to Built-in Data

Tableau comes with a great built-in dataset called "*Sample - Superstore*" that contains mock retail data.

1. Open Tableau Desktop.
2. On the left side of the welcome screen, under Saved Data Sources, click on Sample - Superstore. A blank workspace (Sheet 1) will open.

Step 3: Create Visualization 1 (Sales by Category)

1. On the left pane (the Data pane), find Category. Click and drag it to the Rows shelf at the top.
2. Find Sales (under Measures). Click and drag it to the Columns shelf.
3. Tableau will automatically generate a horizontal bar chart.
4. *Optional:* Click the "Sort Descending" icon on the top toolbar to order the bars from highest to lowest.

Step 4: Create Visualization 2 (Profit Map)

1. At the bottom of the screen, click the icon for New Worksheet (it looks like a small tab with a plus sign).
2. In the Data pane, double-click on State/Province. Tableau will automatically generate a map of the United States with a blue dot on each state.
3. In the Data pane, find Profit. Click and drag it onto the Color square inside the "Marks" card (located to the left of your map).
4. The map will color-code: Blue indicates positive profit, and Orange/Red indicates negative profit (a loss).

Step 5: Tableau Diagnostic Questions

Answer the following in the same Word document you used for Part 1:

4. Based on your first visualization (Sales by Category), which product category generated the highest total sales?

5. Based on your second visualization (Profit Map), identify at least one US State that is operating at a loss (colored orange).

Step 6: Save Tableau Output

1. In Tableau, go to the top menu bar and click **File** > **Print to PDF**.
2. Ensure "Entire Workbook" is selected so both sheets are included.
3. Click OK and save the file as Tableau_Results.pdf.

Part 3: Final Submission Instructions

1. Ensure your Word document containing the answers to the 5 diagnostic questions is complete and save it as a PDF (Answers.pdf).
2. Use a free online PDF merger (like Adobe Acrobat online or standard Mac/Windows tools) to combine your three files:
   - Answers.pdf
   - JupyterLab_Lab_Results.pdf
   - Tableau_Results.pdf

   Name the final combined file YourLastName_DATA300_Week1_Readiness.pdf.
   Submit this single PDF to the course portal in GAP.s

Lab 1: Data Exploration CLO 1, 2, 3

Overview

This lab introduces the foundational practices of descriptive analytics through the exploration, preparation, and visualization of business data using JupyterLab Lab. By the end of this lab, students will be able to identify business questions that can be addressed through descriptive analytics (CLO 1), apply data exploration and visualization techniques to communicate analytical findings (CLO 2), and use computational methods to examine structured datasets and uncover meaningful relationships (CLO 3). These competencies reflect the work of data analysts and business intelligence professionals who transform raw organizational data into actionable insights for decision-making. The knowledge and skills developed in this lab establish the analytical foundation for dashboard development, text mining, predictive modeling, and subsequent applied assignments.

Objective

This hands-on lab focuses on the foundations of descriptive analytics, business problem identification, and advanced initial data exploration. You will interact directly with a raw dataset to execute data quality checks, pivot table aggregations, and multivariate visualizations to build computational problem-solving skills.

The Business Scenario

You are a data analyst for **LuxeGems**, an international jewelry retailer. The purchasing department wants to optimize their diamond buying strategy to ensure competitive pricing while maintaining margin quality. They have provided you with a dataset of over $50,000$ diamonds containing physical attributes (carat weight, cut quality, color grade, clarity rating, physical dimensions) and retail prices in USD ($).
Your goal is to explore this raw data, evaluate data quality, analyze how different attributes impact pricing, and uncover structural relationships that influence strategic inventory acquisition.

Part 1: Accessing and Setting Up JupyterLab

JupyterLab is an open-source, web-based interactive development environment for notebooks, code, and data.

- Open your web browser and navigate to: https://jupyter.org/try-jupyter/lab/
- Create a new notebook by clicking the **Python (Pyodide)** or **Python 3** icon under the **Notebook** section in the Launcher tab.
- Rename your notebook file:
  - Right-click the notebook tab or file name in the left panel -> Select **Rename Notebook.**
  - Name format: YourLastName_Data300_AppliedLab1.ipynb

Part 2: Data Loading and Structural Inspection

Before building analytical models, analysts must inspect data types, confirm dataset dimensions, and check for missing values.

Note for Pyodide Users: Because the Pyodide kernel runs Python inside your browser via WebAssembly, you must install seaborn into the browser session using %pip install seaborn before importing it.

In a new **Code** cell, paste and run the following code block (press Shift + Enter or click the Play button in the notebook toolbar):

```
# 0. Install required library in Pyodide browser environment
```

```
%pip install seaborn

# 1. Import core analytics libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
from pyodide.http import open_url

# 2. Load the built-in 'diamonds' dataset using browser fetch
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/diamonds.csv"
diamonds = pd.read_csv(open_url(url))

# 3. Structural Inspection
print("--- Dataset Information ---")
diamonds.info()

print("\n--- Missing Values Check ---")
print(diamonds.isnull().sum())
print("\n--- Descriptive Statistics (Numeric Features) ---")
display(diamonds.describe().round(2))
```

Part 3: Business Aggregations (Pivot Tables)
To solve business problems, analysts evaluate multi-dimensional relationships. We will construct a **Pivot Table** to examine average retail price grouped by diamond **Cut** (rows) and **Color** (columns).
Context: Diamond color is graded from **D** (colorless/best) to **J** (faint yellow/worst). Diamond cut ranges from **Fair** (worst) to **Ideal** (best).
In a new **Code** cell, paste and run the following code:

```
# Create a Pivot Table to analyze Average Price grouped by Cut and Color
price_pivot = pd.pivot_table(
    data=diamonds,
    values='price',
    index='cut',

    columns='color',
    aggfunc='mean'
)

print("--- Average Diamond Price ($) by Cut Quality and Color Grade ---")
display(price_pivot.round(2))
```

Part 4: Advanced Multivariate Visualizations
Standard scatter plots can become illegible when plotting $50,000+ observations due to point overlap (over-plotting). To address this, we use three specialized visualization techniques:

- **Correlation Heatmaps** to quantify numeric relationships.
- **Violin Plots** to view multi-modal price distributions across cut categories.
- **Hexbin Density Plots** to visualize point density across continuous fields (carat vs. price).

In a new **Code** cell, paste and run the following code:

```
# Set visualization style
sns.set_theme(style="whitegrid")

# 1. Correlation Heatmap
plt.figure(figsize=(10, 8))
numeric_cols = diamonds.select_dtypes(include=[np.number])
corr_matrix = numeric_cols.corr()

sns.heatmap(
    corr_matrix,
    annot=True,
    cmap='coolwarm',
    fmt=".2f",
    linewidths=0.5,
    vmin=-1,
    vmax=1

)
plt.title('Correlation Heatmap of Diamond Attributes', fontsize=14)
plt.show()

# 2. Violin Plot (Price Distribution by Cut Quality)
plt.figure(figsize=(10, 6))
sns.violinplot(
    data=diamonds,
    x='cut',
    y='price',
    hue='cut',
    palette='muted',
    legend=False
)
plt.title('Price Distribution across Diamond Cut Quality Categories', fontsize=14)
plt.xlabel('Cut Quality')
plt.ylabel('Price ($)')

plt.show()

# 3. Hexbin Density Plot (Carat Weight vs. Retail Price)
plt.figure(figsize=(9, 6))
plt.hexbin(
    diamonds['carat'],
    diamonds['price'],
    gridsize=40,
    cmap='Blues',
    mincnt=1
)
cb = plt.colorbar(label='Number of Diamonds in Hexagonal Bin')
plt.title('Density Plot: Carat Weight vs. Price', fontsize=14)
plt.xlabel('Carat Weight')
plt.ylabel('Price ($)')
```

```
plt.show()
```

Part 5: Applied Analytics Follow-up Questions
Create a **Markdown** cell at the bottom of your notebook and answer the following questions in complete sentences:

Section A: Data Quality & Structure

- **Dataset Dimensions & Integrity:**
  - How many total observations (rows) are present in the dataset based on your output from Part 2?
  - Are there any missing (null / NaN) values that require data cleaning before further modeling?

Section B: Business Aggregation & Diagnostic Analytics

- **Pivot Table Price Anomaly:**
  - Examine the Pivot Table in Part 3. One would naturally expect an **Ideal** cut diamond to command a higher price than a **Fair** cut diamond. However, look at color grade **E**. Which cut displays the higher average price in this summary table: **Fair** or **Ideal**?
- **Critical Business Thinking (Confounding Variables):**
  - Why does a **Fair** cut diamond exhibit a higher average price than an **Ideal** cut diamond in this overall summary table? *(Hint: Consider physical volume or weight attributes (carat) that are ignored when aggregating purely by cut and color).*

Section C: Correlation & Graphical Interpretation

- **Correlation Coefficient Analysis:**
  - Review the Correlation Heatmap from Part 4. Which specific numeric attribute exhibits the strongest positive linear correlation (r) with retail price? State the exact numerical coefficient value.

- **Distributional Skewness:**
  - Inspect the Violin Plot from Part 4. Describe the overall shape of the price distributions. Is diamond pricing normally distributed (symmetric bell curve) or heavily right-skewed? Where is the majority of the pricing volume concentrated?

- **Density & Inventory Concentration:**
  - Inspect the Hexbin Density Plot from Part 4. What does the darkest blue cluster indicate regarding LuxeGems' current inventory? Specify both the approximate **Carat weight range** and **Price range ($)** where observation density is highest.

Part 6: Saving and Exporting Your Work
Step 1: Save and Export to PDF in JupyterLab

1. Confirm that all code cells have been executed sequentially and that all output tables, pivot tables, heatmaps, violin plots, and hexbin plots are fully rendered.
2. Confirm that your answered analytical questions are written in a Markdown cell at the bottom of the notebook.
3. In the top JupyterLab menu bar, click **File** -> **Print** (or **Save and Export Notebook As...** -> **HTML**).
4. In your browser's print dialog window:
   - Change the **Destination** to **Save as PDF**.
   - Enable **Background graphics** to preserve colors on heatmaps, violin plots, and density plots.
   - Click **Save** and name your file YourLastName_DATA300_AppliedLab1.pdf.

Step 2: Submit to LMS

1. Log into the university Learning Management System (**GAP**).
2. Navigate to the **DATA 300: Applied Lab 1** drop-box.
3. Upload your saved **PDF file** (YourLastName_DATA300_AppliedLab1.pdf) and click **Submit**.

Grading Expectations

Successful completion of this lab requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Lab 1 Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

Week 2

Learning Materials:

Read

Gordon, M. E. (2023). Business analytics. SAGE Publications Ltd.

- Chapter 5: Communicating Analytical Results
- Chapter 6: Marketing Analytics

Tableau. (n.d.). *What is data visualization? Definition, examples, and learning resources*. https://www.tableau.com/visualization/what-is-data-visualization

Software / Platforms

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Watch

Learn Tableau in 15 minutes and create your first report (FREE sample files). (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=tNLp_JVipoQ

End-to-end Tableau for beginners. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=OmNeLFmyMJQ

Lab 2 Dashboard Creation CLO 1, 2, 3

Overview

This lab develops practical data visualization skills through the creation of interactive dashboards using Tableau. By the end of this lab, students will be able to communicate analytical findings through effective visualizations (CLO 2) while applying descriptive analytics techniques to evaluate business performance (CLO 1) and organize analytical information using computational tools (CLO 3). Interactive dashboards are widely used by business intelligence professionals to monitor organizational performance and support executive decision-making. The experience gained in this lab prepares students to communicate increasingly sophisticated analytical results throughout the course.

Objective: This applied laboratory session engages you in a practical application of data visualization tools. You will manipulate Tableau Desktop to generate a user-centered, interactive business dashboard that visually communicates underlying data trends. This hands-on practice directly reinforces the principles of effective visual data communication and analytical reporting.

The Business Problem: You are an analytics consultant for a nationwide retail superstore. Regional managers are complaining that they cannot easily track their sales trends or identify which products are causing profit losses in their specific states.
Your task is to build a single, interactive "Executive Summary Dashboard." This dashboard must allow regional managers to click on their state and instantly see how their specific sales are trending over time and which product sub-categories are performing the best (or worst).

Part 1: Connecting to the Data

1. Open **Tableau Desktop**.
2. On the left side of the welcome screen, under **Saved Data Sources**, click on **Sample - Superstore**.
3. A blank workspace (Sheet 1) will open. You are now connected to the data.

Part 2: Building the Component Visualizations

A dashboard is made up of individual worksheets. We will build three different charts first.

Step 1: Create the "Monthly Sales Trend" Line Chart

1. At the bottom of the screen, right-click the **Sheet 1** tab and rename it to Monthly Sales Trend.
2. In the Data pane (left side), find **Order Date**. Drag and drop it into the **Columns** shelf at the top.
3. *Important:* Right-click the blue YEAR(Order Date) pill in the Columns shelf. In the menu, select the *second* **Month** option (the one that shows Year and Month, e.g., May 2015). The pill will turn green (Continuous).
4. Drag **Sales** from the Data pane into the **Rows** shelf. You should now see a line chart.

Step 2: Create the "Profitability by State" Map

1. At the bottom of the screen, click the **New Worksheet** icon (the small tab with a plus sign).
2. Right-click the new tab and rename it to Profitability by State.
3. In the Data pane, double-click on **State**. Tableau will automatically generate a map with a dot for each state.
4. On the **Marks** card (to the left of the map), click the dropdown menu that says "Automatic" and change it to **Map** (this creates a filled map).
5. Drag **Profit** from the Data pane and drop it directly onto the **Color** square inside the Marks card.
   (States with positive profit will be blue; negative profit will be orange).

Step 3: Create the "Sales by Sub-Category" Bar Chart

1. Click the **New Worksheet** icon again.
2. Rename this tab to Sales by Sub-Category.
3. Drag **Sub-Category** to the **Rows** shelf.
4. Drag **Sales** to the **Columns** shelf.
5. In the top toolbar, click the **Sort Descending** icon (it looks like a bar chart with a down arrow) to order the products from highest sales to lowest.
6. Drag **Profit** and drop it onto the **Color** square in the Marks card. *(Now the length of the bar shows Sales, but the color shows whether those sales were profitable).*

Part 3: Assembling the Interactive Dashboard

Now we will combine these three sheets into a single interactive dashboard.

1. At the bottom of the screen, click the **New Dashboard** icon (it looks like a square divided into four quadrants, right next to the New Worksheet icon).

2. Under the "Size" menu on the left, change the dropdown from "Range" to **Automatic** so the dashboard fills your screen.
3. On the left side, under "Sheets", you will see the three charts you just built.
4. **Drag and drop** them onto the blank canvas:
   - Drag the **Map** to the top half of the screen.
   - Drag the **Sales Trend** line chart to the bottom left.
   - Drag the **Sub-Category** bar chart to the bottom right.
   - **Adding Interactivity (Crucial Step):** * Click once on your Map chart inside the dashboard to select it (it will get a gray border).
   - In the top right corner of the map's gray border, click the **Filter icon** (it looks like a tiny funnel). It will turn white when activated.
   - *This turns your map into a global filter. Clicking a state will now filter the other two charts.*

Part 4: Applied Analytics Follow-up Questions
Now that your dashboard is fully interactive, use it to answer the following business questions. Type your answers in a separate Word document.

Baseline Exploration:
1. Without clicking on any specific state (looking at the national data), which Sub-Category generates the most overall Sales? Is that sub-category profitable (is the bar blue or orange)?
2. Looking at the national map, name two states that are operating at an overall loss (colored orange).

Interactive Analysis (Using your Dashboard Filter):
3. **Click on the state of Texas** on your map. Notice how the bottom two charts update.
   - Look at the "Sales by Sub-Category" chart for Texas. Which sub-category is causing the most significant financial loss (darkest orange/lowest profit) in Texas?
   - Look at the "Monthly Sales Trend" for Texas. In what month and year did Texas experience a massive, unusual spike in sales?

     Click on the state of California on your map.
   - Compare California's "Sales by Sub-Category" to Texas's. Are the top three best-selling products the same in both states? If not, what are California's top three?

     Critical Thinking (Visual Design): Why is it valuable to color-code the "Sales by Sub-Category" bar chart by *Profit* rather than just coloring it by Sales? What hidden business insight does this dual-axis visual reveal that a standard bar chart would hide?

Part 5: Final Submission Instructions
1. Ensure your Word document containing the answers to the 5 questions is complete and save it as a PDF (Answers_Lab2.pdf).
2. In Tableau, ensure no specific state is selected on your map (click the white background of the map to deselect everything).
3. Go to the top menu bar and click **File** > **Print to PDF**.
4. Under "Print Range", select **Active Sheet** (which should be your Dashboard).
5. Click OK and save the file as Tableau_Dashboard_Lab2.pdf.
6. Combine the two PDFs into a single file named YourLastName_DATA300_AppliedLab2.pdf.
7. Submit this single PDF to the course portal.

Grading Expectations

Successful completion of this lab requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Lab 2 Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

Assignment 1: Data Visualization and Descriptive Analytics Report CLO 1, 2, 3

Assignment 1: Data Visualization and Descriptive Analytics Report

Overview

This assignment applies descriptive analytics and data visualization techniques to communicate meaningful business insights using analytical tools and business datasets. By the end of this assignment, students will be able to describe how descriptive analytics supports organizational decision-making (CLO 1), design effective visualizations that communicate analytical findings (CLO 2), and apply computational methods to organize and analyze business data (CLO 3). These competencies reflect the responsibilities of business analysts who transform complex information into actionable reports for organizational stakeholders. This assignment establishes the analytical communication skills that support all subsequent predictive modeling activities.

Objective

This assignment evaluates your ability to apply descriptive analytics and data visualization principles to unstructured operational data. You will execute an automated Python analytics script in a web-based interactive environment, analyze multi-dimensional statistical summaries and visual figures, and synthesize your findings into a formal, structured business report.

The Business Scenario

You have been hired as an analytics consultant for **Bistro Analytics**, a management consulting firm specializing in restaurant operations optimization. A high-volume client restaurant has provided a comprehensive dataset containing individual transaction records, including total bills, tip amounts, party sizes, dining times (Lunch vs. Dinner), and operational days of the week.

Core Analytics Goal

The restaurant owner seeks data-backed insights to solve two core operational challenges:

- **Labor Scheduling Optimization:** Determining peak revenue windows to strategically schedule service staff.
- **Revenue Expansion & Server Retention:** Understanding tipping dynamics to incentivize upselling and improve server compensation.

Part 1: Running the Analytics Script

You are not required to write original Python code for this assignment. Follow the instructions below to execute the provided computational script in **JupyterLab**, which will render the exact statistical tables and graphics needed for your report.

Execution Steps

- Open your web browser and navigate to **Try JupyterLab**: https://jupyter.org/try-jupyter/lab/.
- Create a new notebook by clicking the **Python (Pyodide)** or **Python 3** icon under the **Notebook** section in the Launcher tab.

- Rename your notebook file using the standard convention: YourLastName_DATA300_Assignment1.ipynb.
- Copy the entire Python code block below, paste it into the first cell, and press **Shift + Enter** (or click the **Play** button in the toolbar).

```
# 0. Install required visualization libraries (required for web-based kernels)
%pip install seaborn pandas matplotlib -q

# 1. Import core libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from pyodide.http import open_url

# 2. Load built-in 'tips' dataset via Pyodide browser fetch wrapper
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv"
df = pd.read_csv(open_url(url))

# 3. Output Table 1: Average Transaction Metrics by Day of Week
print("--- Table 1: Average Transaction Metrics by Day ---")
day_summary = df.groupby('day', observed=True)[['total_bill', 'tip', 'size']].mean().round(2)
day_summary.columns = ['Avg Total Bill ($)', 'Avg Tip ($)', 'Avg Party Size']
display(day_summary)

# 4. Output Table 2: Aggregate Revenue Breakdown by Meal Shift
print("\n--- Table 2: Aggregate Revenue Breakdown by Meal Shift ---")
time_summary = df.groupby('time', observed=True)[['total_bill']].sum().round(2)
time_summary.columns = ['Total Revenue ($)']
display(time_summary)

# Set global Seaborn visual theme
sns.set_theme(style="whitegrid", palette="Set2")

# 5. Figure 1: Total Revenue by Day of Week (Bar Chart)
plt.figure(figsize=(8, 5))
rev_by_day = df.groupby('day', observed=True)['total_bill'].sum().reset_index()
sns.barplot(data=rev_by_day, x='day', y='total_bill', hue='day', legend=False)
plt.title('Figure 1: Total Gross Revenue Generated by Day of Week', fontsize=14,
fontweight='bold')
plt.xlabel('Day of Week')
plt.ylabel('Total Revenue ($)')
plt.tight_layout()
plt.show()

# 6. Figure 2: Tip Distribution across Meal Shifts (Boxplot)
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='time', y='tip', hue='time', legend=False, palette="pastel")
plt.title('Figure 2: Distribution of Server Tip Amounts: Lunch vs. Dinner', fontsize=14,
fontweight='bold')
plt.xlabel('Meal Shift')
plt.ylabel('Tip Amount ($)')
```

```
plt.tight_layout()
plt.show()

# 7. Figure 3: Bivariate Correlation between Bill Size and Tip Amount (Scatter Plot)
plt.figure(figsize=(9, 6))
sns.regplot(
    data=df,
    x='total_bill',
    y='tip',
    scatter_kws={'alpha': 0.6},
    line_kws={'color': 'red', 'linewidth': 2}
)
plt.title('Figure 3: Linear Relationship: Total Bill Size vs. Tip Amount', fontsize=14,
fontweight='bold')
plt.xlabel('Total Bill Amount ($)')
plt.ylabel('Tip Amount ($)')
plt.tight_layout()
plt.show()
```

Part 2: Structure of the Written Analytics Report
After executing the script, review the rendered tables and figures carefully. You will author a formal, professional business report summarizing your diagnostic insights.

Create a Word document (or Markdown file) using the **exact section headings** defined below. Write all responses in complete, polished narrative paragraphs rather than bulleted lists (except where specified for recommendations).

1. Executive Summary *(1 paragraph)*
- State the overarching business objective, the consulting context (**Bistro Analytics**), and the dataset analyzed (tips dataset, N = 244 transactions).
- Summarize the primary operational goal: empowering the restaurant owner to optimize labor scheduling, increase total revenue, and foster server retention through data-driven operational adjustments.

2. Descriptive Data Overview *(1–2 paragraphs)*
- **Day-of-Week Metrics (Table 1):** Analyze the average transaction metrics across days. Identify which day exhibits the highest average total bill ($) and which day yields the highest average tip ($).
- **Shift Revenue Split (Table 2):** Evaluate the total gross revenue generated by **Lunch** versus **Dinner** shifts. Contrast the volume disparity between daytime and evening operations.

3. Visual Insights & Graphical Trends *(2–3 paragraphs)*
- **Gross Revenue Trend (Figure 1):** Interpret the bar chart distribution across the week. Compare weekday operational volume (Thursday/Friday) against weekend demand (Saturday/Sunday). Highlight peak revenue generation days.
- **Tip Distribution & Dispersion (Figure 2):** Compare the central tendencies (median lines) and interquartile ranges (IQR box spread) between Lunch and Dinner. Discuss server earning potential across shifts and address high-value "outliers" (data points extending beyond the upper whiskers).

- **Bivariate Correlation & Upselling Dynamics (Figure 3):** Analyze the slope and trajectory of the OLS regression trendline relating total_bill to tip. Explain what this positive linear relationship implies regarding server upselling initiatives (e.g., promoting high-margin appetizers, premium drinks, and desserts).

4. Strategic Business Recommendations *(2 detailed bullet points)*
Provide two actionable, data-justified operational strategies for the restaurant owner:
- **Recommendation 1 (Staffing & Labor Allocation):** Specify exact days and shifts where the highest-performing service staff should be scheduled to capture peak demand without creating service bottlenecks.
- **Recommendation 2 (Revenue Expansion & Server Training):** Outline a concrete upselling and service training program derived from the bill-to-tip correlation, explaining how higher check averages directly enhance both restaurant margin and server earnings.

Part 3: Submission & Export Instructions
Step 1: Export Your Written Business Report
- Ensure your written report contains all four required sections formatted with the exact headings specified above.
- Export/Save your written document as a PDF named YourLastName_Report.pdf.

Step 2: Export Executed Code Notebook
1. Verify that all Python code cells have run sequentially and that all output tables and Figures 1, 2, and 3 are fully visible.
2. In JupyterLab, go to **File** -> **Print** (or **Save and Export Notebook As...** -> **HTML**).
3. In your browser's print dialog, select **Save as PDF**, enable **Background graphics**, and save the file as YourLastName_Jupyter.pdf.

Step 3: Combine and Submit to LMS
- Merge the two PDF files into a single master document in sequential order (Report first, followed by executed notebook output):
  Master PDF Name: YourLastName_DATA300_Assignment1.pdf
- Log into the university Learning Management System (**GAP**).
- Navigate to the **DATA 300: Assignment 1** drop-box, upload your merged PDF file, and click **Submit**.

Grading Expectations
Successful completion of this assignment requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Assignment 1: Data Visualization and Descriptive Analytics Report Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

Week 3

Learning Materials:

Read

Label Your Data. (n.d.). *Unstructured text in data mining: Unlocking valuable insights in document processing*.
https://labelyourdata.com/articles/unstructured-text-in-data-mining

Kapiche. (n.d.). *Text mining made easy: Definitions, step-by-step tutorial*. https://www.kapiche.com/blog/text-mining

Software / Platforms

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Watch

What is text mining? (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=BeDeHntF68M&t=45s

What is text mining? Explained in 10 minutes. AI-machine learning tutorial. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=V0gxfAWwJSM

Lab 3: Text Mining Application CLO 1, 2, 3

Overview

This lab introduces text mining techniques for extracting meaningful information from unstructured business data. By the end of this lab, students will be able to apply computational methods to discover patterns within textual information (CLO 3) while recognizing how descriptive and predictive analytics support business problem-solving (CLO 1) and communicating analytical findings through appropriate visualization techniques (CLO 2). Text mining is widely applied to customer feedback, social media analysis, and organizational intelligence initiatives. The analytical methods introduced in this lab prepare students for predictive modeling and business intelligence applications later in the course.

Objective

This hands-on laboratory session introduces text mining and natural language processing (NLP) techniques. You will extract structured insights—specifically key phrases and sentiment scores—from unstructured text data using computational algorithms (NLTK and VADER). This exercise builds practical skills for discovering quantitative patterns within unstructured customer feedback.

The Business Scenario

You are an analytics consultant for **OmniTech Solutions**. The customer service department has collected a batch of unstructured text reviews for their flagship tablet, the **OmniPad Pro**. Manually reading thousands of reviews is inefficient and unscalable.

Your task is to implement Python-based Text Mining algorithms to:

- Tokenize text and extract frequent key terms to identify high-volume customer discussion topics.
- Apply a lexicon-based Sentiment Analysis algorithm to quantify customer sentiment on a continuous scale from -1.0 (extremely negative) to +1.0 (extremely positive).
- Translate numerical scores into actionable product engineering recommendations.

Part 1: Accessing and Setting Up JupyterLab

JupyterLab is an open-source, web-based interactive development environment for notebooks, code, and data.

- Open your web browser and navigate to: https://jupyter.org/try-jupyter/lab/
- Create a new notebook by clicking the Python (Pyodide) or Python 3 icon under the Notebook section in the Launcher tab.
- Rename your notebook file:
  - Right-click the notebook tab or file name in the left panel -> Select Rename Notebook...

- Name format: YourLastName_Data300_AppliedLab3.ipynb

Part 2: Environment Setup and Data Loading
In text mining workflows, the initial step involves installing and downloading necessary NLP packages and sentiment lexicons. We will use the NLTK (Natural Language Toolkit) and WordCloud libraries.

Note for Pyodide Users: Because the Pyodide kernel runs Python inside your browser via WebAssembly, you must install external libraries into the browser session using %pip install before importing them.

In a new **Code** cell, paste and run the following code block (press Shift + Enter or click the Play button in the toolbar):

```
# 1. Import core packages
%pip install nltk wordcloud seaborn

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import nltk
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from wordcloud import WordCloud

import io
import os
import zipfile

from pyodide.http import pyfetch

# Helper function to safely fetch binary zip files in browser Python
async def download_nltk_package(category, package_name):
    url =
f"https://raw.githubusercontent.com/nltk/nltk_data/gh-pages/packages/{category}/{package_na
me}.zip"
    target_dir = os.path.expanduser(f"~/nltk_data/{category}")
    os.makedirs(target_dir, exist_ok=True)

    response = await pyfetch(url)
    data = await response.bytes()

    with zipfile.ZipFile(io.BytesIO(data)) as z:
        z.extractall(target_dir)

# 2. Download required NLTK datasets and lexicons
await download_nltk_package('sentiment', 'vader_lexicon')
await download_nltk_package('tokenizers', 'punkt')
await download_nltk_package('tokenizers', 'punkt_tab')
await download_nltk_package('corpora', 'stopwords')
```

```
# 3. Load unstructured customer review dataset
data = {
    'Review_ID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Review_Text': [
        "Absolutely love the new OmniPad! The battery life is amazing and the screen is beautiful.",
        "Terrible experience. The screen cracked after one week and customer service was completely useless.",
        "It is an okay tablet. Nothing special, but it gets the job done for basic web browsing.",
        "The battery life is incredibly poor. I have to charge it twice a day. Very disappointed.",
        "Fantastic device! Fast processor, great camera, and very lightweight. Highly recommend.",
        "Overpriced garbage. The software is laggy and it crashes constantly.",
        "A solid upgrade from the previous generation. The stylus integration is perfect for my artwork.",
        "I hate the new update. It ruined the user interface and made everything slower.",
        "Decent value for the price. The battery is fine, but the camera could be better.",
        "Best purchase I've made this year. Beautiful display and amazing battery life!"
    ]
}

df = pd.DataFrame(data)
print("--- Raw Customer Reviews Preview ---")
display(df.head())
```

Part 3: Text Preprocessing & Key Phrase Extraction
Algorithms require unstructured text to be preprocessed before extracting statistical patterns. We must tokenize the text (split sentences into individual words) and filter out non-informative "stop words" (e.g., "the", "is", "and").

In a new **Code** cell, paste and run the following code to construct a Word Cloud and generate a word frequency distribution:

```
# 1. Combine all review texts into a single string
all_text = " ".join(review for review in df['Review_Text']).lower()
# 2. Tokenize text and configure stop word filter
stop_words = set(stopwords.words('english'))
# Add custom domain-specific stop words that do not add distinct signal
stop_words.update(['omnipad', 'tablet', 'device'])

words = word_tokenize(all_text)
cleaned_words = [word for word in words if word.isalpha() and word not in stop_words]

# 3. Generate and plot Word Cloud visualization
wordcloud = WordCloud(
    width=800,
    height=400,
    background_color='white',
    colormap='viridis'
).generate(" ".join(cleaned_words))
```

```
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis("off")
plt.title("Key Phrase Extraction: OmniPad Reviews Word Cloud\n", fontsize=16)
plt.show()

# 4. Extract Top 5 Most Frequent Keywords
freq_dist = nltk.FreqDist(cleaned_words)
print("\n--- Top 5 Most Frequent Keywords ---")
for word, frequency in freq_dist.most_common(5):
    print(f"Keyword: '{word}' | Frequency Count: {frequency}")
```

Part 4: Structured Sentiment Analysis (VADER)
We will apply **VADER** (*Valence Aware Dictionary and sEntiment Reasoner*), a rule-based sentiment model optimized for product reviews and short social media texts. VADER calculates a normalized compound sentiment score S in [-1.0, +1.0] for each review:

- **Positive Sentiment:** S >= 0.05
- **Neutral Sentiment:** -0.05 < S < 0.05
- **Negative Sentiment:** S <= -0.05

In a new **Code** cell, paste and run the following code:

```
import io
import os
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import nltk
from pyodide.http import pyfetch

# --- 1. Download VADER Lexicon via Pyodide Fetch ---
async def setup_vader():
    url =
"https://raw.githubusercontent.com/nltk/nltk_data/gh-pages/packages/sentiment/vader_lexicon.
zip"
    target_dir = os.path.expanduser("~/nltk_data/sentiment")
    os.makedirs(target_dir, exist_ok=True)
    # Fetch binary bytes cleanly
    response = await pyfetch(url)
    content = await response.bytes()

    # Save the raw zip file where NLTK expects it
    with open(os.path.join(target_dir, "vader_lexicon.zip"), "wb") as f:
        f.write(content)

# Execute async download
await setup_vader()

# --- 2. Initialize VADER Analyzer ---
from nltk.sentiment.vader import SentimentIntensityAnalyzer
```

```
analyzer = SentimentIntensityAnalyzer()

# --- 3. Customer Review Data ---
data = {
    'Review_ID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Review_Text': [
        "Absolutely love the new OmniPad! The battery life is amazing and the screen is
beautiful.",
        "Terrible experience. The screen cracked after one week and customer service was
completely useless.",
        "It is an okay tablet. Nothing special, but it gets the job done for basic web browsing.",
        "The battery life is incredibly poor. I have to charge it twice a day. Very disappointed.",
        "Fantastic device! Fast processor, great camera, and very lightweight. Highly
recommend.",
        "Overpriced garbage. The software is laggy and it crashes constantly.",
        "A solid upgrade from the previous generation. The stylus integration is perfect for my
artwork.",
        "I hate the new update. It ruined the user interface and made everything slower.",
        "Decent value for the price. The battery is fine, but the camera could be better.",
        "Best purchase I've made this year. Beautiful display and amazing battery life!"
    ]
}
df = pd.DataFrame(data)

# --- 4. Sentiment Analysis & Categorization ---
df['Compound_Score'] = df['Review_Text'].apply(
    lambda x: analyzer.polarity_scores(x)['compound']
)

def categorize_sentiment(score):
    if score >= 0.05:
        return 'Positive'
    elif score <= -0.05:
        return 'Negative'
    else:
        return 'Neutral'

df['Sentiment_Label'] = df['Compound_Score'].apply(categorize_sentiment)

print("--- Structured Sentiment Results ---")
display(df[['Review_ID', 'Review_Text', 'Compound_Score', 'Sentiment_Label']])

# --- 5. Plot Sentiment Distribution ---
plt.figure(figsize=(8, 5))
sns.set_theme(style="whitegrid")
sns.countplot(

    data=df,
    x='Sentiment_Label',
    hue='Sentiment_Label',
    palette={'Positive': 'green', 'Neutral': 'gray', 'Negative': 'red'},
```

```
    legend=False
)
plt.title('Distribution of Customer Sentiment Categories', fontsize=16)
plt.xlabel('Sentiment Class')
plt.ylabel('Review Count')
plt.show()
```

Part 5: Applied Analytics Follow-up Questions
Create a **Markdown** cell at the bottom of your notebook and answer the following questions in complete sentences:

Section A: Text Preprocessing & Key Phrase Extraction

- **Stop Word Removal Rationale:**
  - Define what a "stop word" is in natural language processing. From a data analytics perspective, why is removing stop words necessary before calculating word frequency distributions?

- **Key Phrase Identification:**
  - Examine the Word Cloud and Top 5 Keyword output from Part 3. What specific hardware component or product attribute of the OmniPad Pro is discussed most frequently by customers?

Section B: Sentiment Metric Diagnostic

Compound Score Evaluation:

- Locate Review ID 6 (*"Overpriced garbage. The software is laggy and it crashes constantly."*) in your Part 4 output table. State the exact numerical Compound_Score assigned to this review.
- Does this calculated numerical score align with human qualitative interpretation of the text?

Neutral Threshold Analysis:

- Locate Review ID 3 (*"It is an okay tablet. Nothing special, but it gets the job done for basic web browsing."*). Explain why the VADER algorithm classified this review as **Neutral** based on its mathematical scoring bounds ($-0.05 < S < 0.05$).

Section C: Business Decision Support
Strategic Product Recommendation:

- Synthesize the sentiment distribution chart and extracted key phrases. As an analytics consultant for OmniTech Solutions, what is one actionable recommendation you would submit to the hardware or software engineering teams for the next product iteration? Justify your recommendation using specific data points from your notebook outputs.

Part 6: Saving and Exporting Your Work
Step 1: Save and Export to PDF in JupyterLab

1. Confirm that all code cells have been executed sequentially and that all output previews, word clouds, sentiment summary tables, and bar plots are fully rendered.
2. Confirm that your analytical responses are written in a Markdown cell at the bottom of the notebook.
3. In the top JupyterLab menu bar, click **File** -> **Print** (or **Save and Export Notebook As...** -> **HTML**).

4. In your browser's print dialog window:
   - Change the **Destination** to **Save as PDF**.
   - Enable **Background graphics** to preserve colors on word clouds and sentiment bar charts.
   - Click **Save** and name your file YourLastName_DATA300_AppliedLab3.pdf.

Step 2: Submit to LMS

1. Log into the university Learning Management System (**GAP**).
2. Navigate to the **DATA 300: Applied Lab 3** drop-box.
3. Upload your saved **PDF file** (YourLastName_DATA300_AppliedLab3.pdf) and click **Submit**.

Grading Expectations

Successful completion of this lab requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Lab 3 Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

SM Activity CLO 1, 2, 3

Socratic Metric (SM) Assignment Instructions

This activity is designed to support student learning through verbal explanation, reflection, and revision. The process emphasizes developing conceptual understanding by articulating ideas clearly and responding to feedback.

Getting Started

- Review the Socratic Metric (SM) Student Guide to become familiar with system expectations and functionality.
- Copy the SM Assignment link provided by the instructor in the online course.
- Open the Socratic Metric Assignment Pop Up (indicated by the puzzle icon)
- Paste the link from the instructor in the pop up to begin

Assignment Process

- Read the Question Aloud
  - Speak clearly and at a natural pace.
  - Speech should be sufficiently clear for the system to capture at least 70% accuracy.
- Provide a Verbal Response
  - Explain the concept using your own words.
  - Respond using natural speech, as if explaining the idea in a real-world conversation.
- Use of Learning Resources
  - Reference textbooks or notes as needed.
  - Do not read directly from a script or AI-generated text.
- Review Feedback & Revising Responses
  - Read the system-generated feedback aloud after each submission.
  - Revise each response up to four times.

Submission Requirements

- Upon completion of the final question, select “Finish Assessment.”

- Remain on the page until the “Assignment Complete” screen appears to ensure successful submission.

Grading Criteria
The grading rubric for this assignment evaluates the overall quality of student responses, including the accuracy and relevance of information, effective use of course materials, and the depth of critical thinking demonstrated. View the Socratic Metric grading rubric, and become familiar with the expectations prior to beginning the assignment.

Week 4

Learning Materials:

Read:
Gordon, M. E. (2023). Business analytics. SAGE Publications Ltd.
- Chapter 9: Financial Analytics

TransImpact. (n.d.). *Forecasting methods: The complete guide*.
https://transimpact.com/blog/forecasting-methods/

Software / Platforms
Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Watch
What is time series analysis? (n.d.). [Video]. YouTube.
https://www.youtube.com/watch?v=GE3JOFwTWVM
Forecasting in Excel: MUST-KNOW for any analyst. (n.d.). [Video]. YouTube.
https://www.youtube.com/watch?v=6rqhMnOgQnU

Lab 4 Forecasting CLO 1, 2, 5

Overview
This lab introduces forecasting techniques used to predict future business outcomes from historical data. By the end of this lab, students will be able to apply forecasting methods to business problems (CLO 1), communicate forecasting results using appropriate analytical tools (CLO 2), and evaluate how predictive models support managerial decision-making within business intelligence environments (CLO 5). Forecasting is an essential analytical capability used across finance, operations, healthcare, retail, and supply chain management. The skills developed in this lab establish the predictive foundation for regression modeling and simulation activities that follow.

Objective
In DATA 200, students learned the basics of time series forecasting by fitting an ARIMA model to a complete dataset. In this lab, you will learn professional standards for predictive modeling. You will perform statistical tests for stationarity, split your time series into Training and Testing sets to prevent over-fitting, fit a Seasonal ARIMA (SARIMA) model, and quantitatively evaluate model forecast accuracy using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

The Analytical Scenario
You are a data scientist working for **Global Climate Analytics**. You have been tasked with predicting future atmospheric CO_2 levels based on historical data collected at the Mauna Loa Observatory.

Before presenting your forecasting model to environmental policymakers, you must mathematically prove that your model is accurate. To do this, you cannot simply forecast an unknown future; you must withhold (hide) a portion of recent historical data from your model, generate forecasts for those hidden dates, and evaluate the model's accuracy against actual historical observations.

Part 1: Accessing and Setting Up JupyterLab
JupyterLab is an open-source, web-based interactive development environment for notebooks, code, and data.

- Open your web browser and navigate to: https://jupyter.org/try-jupyter/lab/
- Create a new notebook by clicking the **Python (Pyodide)** or **Python 3** icon under the **Notebook** section in the Launcher tab.
- Rename your notebook file:
  - Right-click the notebook tab or file name in the left panel -> Select **Rename Notebook.**
  - Name format: YourLastName_Data300_AppliedLab4.ipynb

Part 2: Loading Data and Testing for Stationarity
A core assumption of ARIMA modeling is that time series data must be stationary (having constant mean and variance over time). We evaluate stationarity mathematically using the **Augmented Dickey-Fuller (**ADF**)** test.

Note for Pyodide Users: Because the Pyodide kernel runs Python inside your browser via WebAssembly, you must install external libraries (statsmodels and scikit-learn) into the browser session using %pip install before importing them.

In a new **Code** cell, paste and run the following code block (press Shift + Enter or click the Play button in the notebook toolbar):

```
# 0. Install required time series and machine learning packages in Pyodide environment
%pip install statsmodels scikit-learn seaborn matplotlib

# 1. Import core packages
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from statsmodels.tsa.stattools import adfuller
from sklearn.metrics import mean_squared_error, mean_absolute_error
import warnings
warnings.filterwarnings("ignore")

# 2. Load the built-in CO2 dataset
dataset = sm.datasets.co2.load_pandas()

df = dataset.data

# 3. Data Cleaning: Resample weekly data to monthly averages and fill missing values
df = df.resample('ME').mean().ffill()

# 4. Exploratory Time Series Plot
plt.figure(figsize=(10, 4))
```

```
plt.plot(df.index, df['co2'], color='green')
plt.title('Monthly CO2 Levels (Mauna Loa Observatory)')
plt.ylabel('CO2 (ppm)')
plt.xlabel('Date')
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()

# 5. Stationarity Test: Augmented Dickey-Fuller (ADF)
print("--- Augmented Dickey-Fuller (ADF) Test Results ---")
adf_result = adfuller(df['co2'])
print(f"ADF Statistic: {adf_result[0]:.4f}")
print(f"p-value: {adf_result[1]:.4f}")
if adf_result[1] > 0.05:
    print("Conclusion: The p-value > 0.05. The data is NOT stationary (it contains a
non-stationary trend).")
else:
    print("Conclusion: The p-value <= 0.05. The data IS stationary.")
```

Part 3: Train / Test Data Splitting
To evaluate model generalizability, we partition our temporal dataset sequentially. The model will learn parameters from the **Train set** (80% of historical data), and we will evaluate its predictions against the **Test set** (20% hidden holdout data).

In a new **Code** cell, paste and run the following code:

```
# 1. Sequential Train/Test Split (80% Training, 20% Testing)
train_size = int(len(df) * 0.8)
train_data, test_data = df.iloc[:train_size], df.iloc[train_size:]

print(f"Total Observations: {len(df)}")
print(f"Training Horizon:   {len(train_data)} months")
print(f"Testing Horizon:    {len(test_data)} months")

# 2. Visualize Train vs. Test Split
plt.figure(figsize=(10, 4))
plt.plot(train_data.index, train_data['co2'], label='Train Data (80%)', color='blue')
plt.plot(test_data.index, test_data['co2'], label='Test Data (20%) - Hidden Holdout',
color='orange')
plt.title('Sequential Train / Test Partitioning')
plt.ylabel('CO2 (ppm)')
plt.xlabel('Date')
plt.legend(loc='upper left')
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```

Part 4: SARIMA Modeling & Quantitative Accuracy Evaluation
Because the time series exhibits both an overall upward trend and strong 12-month annual seasonality, we fit a Seasonal ARIMA (SARIMA) model:

$$\text{SARIMA}(p, d, q) \times (P, D, Q)_s$$

We fit the model exclusively on the training set (blue series), forecast over the duration of the testing period (orange series), and evaluate error metrics (RMSE and MAE).

In a new **Code** cell, paste and run the following code:

```
# 1. Fit SARIMA model on TRAINING data ONLY
# Order = (p=1, d=1, q=1) | Seasonal Order = (P=1, D=1, Q=1, s=12 months)
print("Fitting the SARIMA Model (This may take 10-20 seconds)...")
model = sm.tsa.statespace.SARIMAX(
    train_data['co2'],
    order=(1, 1, 1),
    seasonal_order=(1, 1, 1, 12)
)
sarima_result = model.fit(disp=False)

# 2. Forecast across the exact temporal span of the TEST set
forecast_steps = len(test_data)
predictions = sarima_result.get_forecast(steps=forecast_steps)
pred_mean = predictions.predicted_mean

# 3. Calculate Quantitative Accuracy Metrics
rmse = np.sqrt(mean_squared_error(test_data['co2'], pred_mean))
mae = mean_absolute_error(test_data['co2'], pred_mean)

print("\n--- Out-of-Sample Model Evaluation Metrics ---")
print(f"Root Mean Squared Error (RMSE): {rmse:.2f} ppm")
print(f"Mean Absolute Error (MAE):      {mae:.2f} ppm")

# 4. Visual Validation: Actual Test Observations vs. Model Predictions
plt.figure(figsize=(12, 6))
plt.plot(train_data.index[-50:], train_data['co2'][-50:], label='Recent Train Data', color='blue')

plt.plot(test_data.index, test_data['co2'], label='Actual Test Data', color='orange')
plt.plot(test_data.index, pred_mean, label='SARIMA Out-of-Sample Forecast', color='red',
linestyle='--')

plt.title('SARIMA Model Performance on Hidden Test Period')
plt.ylabel('CO2 Levels (ppm)')
plt.xlabel('Date')
plt.legend(loc='upper left')
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```

Part 5: Applied Analytics Follow-up Questions
Create a **Markdown** cell at the bottom of your notebook and answer the following questions in complete, clear sentences:

- **Stationarity & Differencing (ADF Test):**
  - Examine your output from Part 2. What was the exact numerical p-value obtained from the Augmented Dickey-Fuller (ADF) test?

- Based on this p-value, why was it necessary to apply first-order differencing (setting the non-seasonal integration parameter d = 1) in our SARIMA specification rather than fitting an undifferenced ARMA model?

- **Train/Test Partitioning Rationale:**
- In Part 3, we withheld the final 20% of historical data from the estimation algorithm during training.
- In a real-world business or environmental science consulting scenario, why is this sequential Train/Test validation strategy far more rigorous than training a model on 100% of historical data at once? What risk does it mitigate?

- **Quantitative Error Metric Evaluation (**MAE**):**
- Review the out-of-sample error metrics printed in Part 4. State your exact calculated Mean Absolute Error (MAE) value.
- Given that baseline atmospheric CO_2 concentration levels in this dataset average approximately 350+ parts per million (ppm), evaluate this error rate. Would you consider this prediction error rate to be excellent, acceptable, or poor? Justify your answer.

- **Visual Forecast Diagnostics:**
- Inspect the final comparative time series plot in Part 4. The red dashed line represents the model's out-of-sample predictions, while the solid orange line represents actual historical observations.
- Did the SARIMA model successfully replicate both the overall trajectory and the intra-annual seasonal "zig-zag" oscillation pattern of atmospheric CO_2?

Part 6: Saving and Exporting Your Work
Step 1: Save and Export to PDF in JupyterLab

1. Confirm that all code cells have been executed sequentially and that all output previews, ADF test statistics, train/test split plots, error metrics, and validation figures are fully rendered.
2. Confirm that your analytical responses are written in a Markdown cell at the bottom of the notebook.
3. In the top JupyterLab menu bar, click **File** -> **Print** (or **Save and Export Notebook As...** -> **HTML**).
4. In your browser's print dialog window:
   - Change the **Destination** to **Save as PDF**.
   - Enable **Background graphics** to preserve line colors and shaded graph elements.
   - Click **Save** and name your file YourLastName_DATA300_AppliedLab4.pdf.

Step 2: Submit to LMS

1. Log into the university Learning Management System (**GAP**).
2. Navigate to the **DATA 300: Applied Lab 4** drop-box.
3. Upload your saved **PDF file** (YourLastName_DATA300_AppliedLab4.pdf) and click **Submit**.

Grading Expectations

Successful completion of this lab requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Lab 4 Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

MIDTERM EXAM - CLO 1, 2, 3, and 4

Week 5

Learning Materials

Read:

Gordon, M. E. (2023). Business analytics. SAGE Publications Ltd.

- Chapter 8: Operational Analytics

Google for Developers. (n.d.). *Logistic regression*. https://developers.google.com/machine-learning/crash-course/logistic-regression

Software / Platforms

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Watch:

Data Science Wednesday. (n.d.). *The fundamentals of predictive analytics* [Video]. YouTube. https://www.youtube.com/watch?v=4y6fUC56KPw

Lab 5: Predictive Linear Regression Modeling CLO 4, 5

Overview

This lab applies multiple linear regression techniques to develop predictive models using real-world business data. By the end of this lab, students will be able to construct predictive regression models to project future outcomes (CLO 4) and evaluate the implications of analytical results within business intelligence contexts (CLO 5). These predictive modeling techniques are routinely applied by data scientists and business analysts to improve forecasting accuracy and support strategic planning. The competencies developed in this lab prepare students for advanced simulation models and comprehensive business intelligence case analyses.

Objective: In this lab, we transition to predictive modeling using the Machine Learning workflow. You will use Python's scikit-learn library to partition data into training and testing sets, fit a multiple linear regression model, evaluate its out-of-sample predictive accuracy using strict metrics (RMSE, MAE, R^2), and diagnose model health through residual analysis.

The Business Scenario

You are a data scientist for an automotive analytics consulting firm. A major car manufacturer wants to optimize their upcoming vehicle designs to maximize fuel efficiency and meet new regulatory standards.

Using historical data, your task is to build a predictive Multiple Linear Regression model that can accurately forecast a vehicle's Miles Per Gallon (MPG) based purely on its engineering specifications (weight, horsepower, displacement, cylinders, and acceleration). You must also prove that your model can generalize to new vehicles it has never seen before.

Part 1: Setting up the JupyterLab Environment

- Open your web browser and go to https://jupyter.org/try-jupyter/lab/.
- Once the workspace loads, you will see the **Launcher** tab in the main area. Under the **Notebook** section, click on **Python 3 (ipykernel)** or **Pyodide** to create a new notebook.
- Rename your notebook file:
  - Go to the top menu bar and click **File** > **Rename Notebook...** (or right-click Untitled.ipynb in the left file browser tab).
  - Rename the notebook to: YourLastName_Data300_AppliedLab5.ipynb.

Part 2: Loading Data and Machine Learning Preprocessing

In predictive analytics, we separate our data into features (X variables) and the target (y variable). We also drop any missing values to ensure the algorithm trains smoothly.

- Click inside the first empty code cell in your notebook.
- Copy and paste the code below into the cell.
- Click the **Play (▶)** button on the toolbar (or press **Shift + Enter**) to run the cell.

```
# Install required libraries if missing in the current Jupyter kernel
%pip install -q seaborn scikit-learn matplotlib pandas numpy

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
# 1. Load the 'mpg' dataset
# Try Jupyter / JupyterLite runs in Pyodide (WASM browser sandbox), which
# restricts standard socket HTTP calls used by `sns.load_dataset()`.
# We use a browser-compatible fetch method with fallback:
url = 'https://raw.githubusercontent.com/mwaskom/seaborn-data/master/mpg.csv'

try:
    import pyodide.http
    df = pd.read_csv(pyodide.http.open_url(url))
except (ImportError, ModuleNotFoundError):
    df = pd.read_csv(url)

# 2. Data Cleaning: Drop rows with missing values
df = df.dropna()

# 3. Define Features (X) and Target (y)
features = ['cylinders', 'displacement', 'horsepower', 'weight', 'acceleration']
```

```
X = df[features]
y = df['mpg']

print("--- Dataset Snapshot ---")
display(df[features + ['mpg']].head())
print(f"\nTotal viable vehicles in dataset: {len(df)}")
```

Part 3: Train/Test Split & Model Training

To prove a model actually predicts rather than just memorizes the data, we must split the dataset. We train the model on 80% of the data, and then ask it to predict the remaining 20% that we hid from it.

- Add a new code cell by clicking the **+ (Insert a cell below)** button on the notebook toolbar or pressing **B** in command mode.
- Paste the following code into the new cell and run it (**Shift + Enter**):

```
# 1. Split the data (80% Training, 20% Testing)
# random_state=42 ensures we all get the exact same random split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42)

print(f"Training on {len(X_train)} vehicles...")
print(f"Testing on {len(X_test)} vehicles (Hidden during training)...\n")

# 2. Initialize and Train the Linear Regression Model
model = LinearRegression()
model.fit(X_train, y_train)

# 3. Extract the calculated coefficients (The business impact)
print("--- Model Coefficients (Impact on MPG) ---")
coefficients = pd.DataFrame({'Feature': features, 'Coefficient': model.coef_})
display(coefficients)
```

Part 4: Out-of-Sample Prediction and Evaluation

Now we give the model the engineering specs (X_test) of the hidden 20% of vehicles and ask it to predict their MPG. We then compare its predictions to the actual known MPG (y_test) to grade its performance.

- Click the **+** toolbar button to add a new code cell.
- Paste the code below and press **Shift + Enter**:

```
# 1. Generate predictions on the hidden TEST set
y_pred = model.predict(X_test)

# 2. Calculate Predictive Evaluation Metrics
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
mae = mean_absolute_error(y_test, y_pred)
r2_test = r2_score(y_test, y_pred)
```

```
print("--- Predictive Performance on TEST Data ---")
print(f"Root Mean Squared Error (RMSE): {rmse:.2f} MPG")
print(f"Mean Absolute Error (MAE):      {mae:.2f} MPG")
print(f"Test R-Squared (R2):            {r2_test:.4f}")

# 3. Visualizing Actual vs. Predicted Values
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.7, color='blue')
# Add a perfect prediction line (y = x)
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--', lw=2)
plt.title('Predictive Accuracy: Actual MPG vs. Predicted MPG')
plt.xlabel('Actual MPG (True Values)')
plt.ylabel('Predicted MPG (Model Guesses)')
plt.show()
```

Part 5: Advanced Residual Diagnostics

A core assumption of advanced regression is **Homoscedasticity**—meaning our errors (residuals) should be randomly scattered. If our model is systematically over-predicting high-MPG cars and under-predicting low-MPG cars, our residuals will show a pattern, meaning our model is flawed.

- Click the **+** toolbar button to add a new cell.
- Paste and execute the following code:

```
# Calculate Residuals (Actual - Predicted)
residuals = y_test - y_pred

# Plot Residuals
plt.figure(figsize=(8, 6))
plt.scatter(y_pred, residuals, alpha=0.7, color='purple')
plt.axhline(y=0, color='r', linestyle='--')
plt.title('Residual Diagnostic Plot (Checking for Homoscedasticity)')
plt.xlabel('Predicted MPG')
plt.ylabel('Residual Error (Actual - Predicted)')
plt.show()
```

Part 6: Applied Analytics Follow-up Questions

Add a **Markdown Cell** at the bottom of your Jupyter notebook (click +, then in the toolbar dropdown menu that currently says **Code**, switch it to **Markdown**). Type/paste your answers directly into the notebook cell and run it (**Shift + Enter**) to render formatted text. Alternatively, complete your answers in a separate Word document if specified by your instructor.

Answer the following based on your outputs:

1. **Model Coefficients (Business Impact):** Look at the Model Coefficients table in Part 3. The coefficient for weight is approximately -0.005. Explain what this number means in plain

English regarding vehicle design. *(Hint: For every 1-pound increase in weight, what happens to the MPG? What about a 1,000-pound increase?)*

2. **Predictive Performance:** Look at the Test R^2 value generated in Part 4. State the value. Because this was calculated on the Test data (data the model never saw during training), what does this specific percentage tell us about the model's true real-world predictive power?
3. **Error Analysis (MAE):** State your Mean Absolute Error (MAE) from Part 4. In a business meeting with automotive engineers, how would you interpret this metric? *(e.g., "When our model predicts a vehicle's MPG, it is typically off by about [X] miles per gallon.")*
4. **Visual Accuracy:** Look at the "Actual vs. Predicted" scatter plot. The red dashed line represents perfect predictions. Are the blue dots tightly clustered around the red line, or widely scattered? Does the model seem to struggle more with predicting low-MPG cars (around 15) or high-MPG cars (around 35+)?
5. **Residual Diagnostics:** Look at the final "Residual Diagnostic Plot." In a perfect model, these purple dots should look like a random cloud of dust centered around the red zero-line. Do you see a random cloud, or do you notice a "U-shape" or widening pattern? If there is a pattern (heteroscedasticity), it suggests our linear model might be missing something (like a non-linear relationship). Based on the plot, what is your conclusion?

Part 7: Final Submission Instructions

- Ensure all code cells have been executed sequentially and all plots/tables are clearly displayed.
- Save your notebook: Click **File** > **Save Notebook** (or press **Ctrl + S** / **Cmd + S**).
- Export your submission:
  - Go to **File** > **Print...** in the top menu bar.
  - In the browser print window, select **Save as PDF** as the printer destination.
  - Name your file YourLastName_DATA300_AppliedLab5.pdf.
    *(Note: Alternatively, if requested by your instructor, you can go to **File** > **Save and Export Notebook As...** > **HTML** file via **File** > **Download**).*
- Submit the saved PDF (or specified file format) to the course learning portal.

Grading Expectations

Successful completion of this lab requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Lab 5 Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

Assignment 2 Predictive Data Mining Report CLO 1, 2, 3, 4

Overview

This assignment applies predictive data mining techniques to identify meaningful patterns and relationships within business data. By the end of this assignment, students will be able to

describe predictive analytical approaches for solving organizational challenges (CLO 1), communicate analytical findings through appropriate visualization techniques (CLO 2), apply computational methods to structured and unstructured datasets (CLO 3), and evaluate predictive model performance to support business decisions (CLO 4). Data mining is a core competency in business intelligence and predictive analytics across virtually every industry. The experience gained through this assignment prepares students for advanced predictive modeling and simulation techniques.

Objective: This assignment challenges you to implement data mining and regression techniques to discover predictive patterns in a complex dataset. You will extract actionable insights from unstructured text data using Natural Language Processing (NLP) and combine them with structured numerical data to build a predictive forecasting model. Then, synthesize your findings into a professional business report.

The Business Scenario
You are a lead data miner for "UrbanTrend Real Estate Analytics." Your firm wants to build an automated algorithm to forecast the market value (Price) of homes.

You have access to structured data (Square Footage, Number of Bedrooms, Age of the Home) and unstructured data (the Real Estate Agent's written description of the property). Historically, analysts have ignored the text descriptions because algorithms cannot easily read sentences.

The Task: Use Text Mining to convert the unstructured agent notes into a mathematical "Text Appeal Score" (using Sentiment Analysis). Then, build a Multiple Linear Regression model that combines the structured house specs with your newly mined text score to predict the final sale price.

Part 1: Setting up the JupyterLab Environment
You do not need to write the Python code for this analysis from scratch. Follow the instructions to run the provided scripts, which will generate the data, visual EDA, text mining features, and the final predictive model.

- Open your web browser and go to https://jupyter.org/try-jupyter/lab/.
- Once the workspace loads, you will see the **Launcher** tab. Under the **Notebook** section, click on **Python 3 (ipykernel)** or **Pyodide** to create a new notebook.
- Rename your notebook file:
  - Go to the top menu bar and click **File** > **Rename Notebook...** (or right-click Untitled.ipynb in the left file browser tab).
  - Rename the notebook to: YourLastName_DATA300_Assignment2.ipynb.

Part 2: Data Generation and Exploratory Data Analysis (EDA)

- Click inside the first empty code cell in your notebook.
- Copy and paste the exact code block below.
- Click the **Play (▶)** button on the toolbar (or press **Shift + Enter**) to execute the cell. This installs required libraries, generates our complex real estate dataset, and explores baseline

numerical trends.

```
# Install required libraries in the active Jupyter kernel
%pip install -q seaborn scikit-learn matplotlib pandas numpy nltk

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import nltk

# 1. Simulate a Complex Real Estate Dataset
np.random.seed(101)
n = 500

sqft = np.random.normal(2200, 600, n)
age = np.random.uniform(1, 60, n)
beds = np.random.randint(2, 6, n)

# Baseline Price Calculation
price = (sqft * 150) - (age * 800) + (beds * 15000)

# Injecting Unstructured Text (Agent Descriptions)
descriptions = []
for i in range(n):
    rand_val = np.random.rand()
    if rand_val > 0.8:
        descriptions.append("Absolutely stunning! Fully renovated with luxury finishes, granite,
and beautiful hardwood. A dream home.")
        price[i] += 45000  # Premium for luxury
    elif rand_val < 0.2:
        descriptions.append("Needs significant work. Major fixer-upper. Roof is old and interior
requires complete cosmetic updates. Sold as-is.")
        price[i] -= 35000  # Penalty for needing repairs
    else:

        descriptions.append("Nice family home in a quiet neighborhood. Standard features, well
maintained. Ready to move in.")

# Add market noise
price += np.random.normal(0, 15000, n)

# Compile into DataFrame
df = pd.DataFrame({'SqFt': sqft, 'Bedrooms': beds, 'Age': age, 'Description': descriptions,
'Price': price})

print("--- Step 1: Raw Data Snapshot (Structured & Unstructured) ---")
display(df.head())

# 2. Exploratory Data Analysis (EDA)
plt.figure(figsize=(14, 5))
```

```
plt.subplot(1, 2, 1)
sns.scatterplot(data=df, x='SqFt', y='Price', alpha=0.6, color='blue')
plt.title('EDA: Square Footage vs. Price')

plt.subplot(1, 2, 2)

sns.boxplot(data=df, x='Bedrooms', y='Price', palette='Set2')
plt.title('EDA: Price Distribution by Bedrooms')
plt.tight_layout()
plt.show()
```

Part 3: Text Mining & Predictive Regression

- Add a new code cell by clicking the **+ (Insert a cell below)** toolbar icon or pressing **B** in command mode.
- Copy and paste the code below into the cell.
- Run the cell (**Shift + Enter**). This will mine the unstructured text for a "Sentiment Score" and feed it into a predictive Machine Learning model.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# 1. Text Mining: Extracting Sentiment from Unstructured Text
# Initialize NLP Lexicon with browser sandbox compatibility
try:
    nltk.download('vader_lexicon', quiet=True)
    from nltk.sentiment.vader import SentimentIntensityAnalyzer
    analyzer = SentimentIntensityAnalyzer()
except Exception:
    # WebAssembly/Browser sandbox fallback if socket requests are restricted
    class SimpleSentimentAnalyzer:
        def polarity_scores(self, text):
            t = text.lower()
            if 'stunning' in t or 'luxury' in t: return {'compound': 0.85}
            elif 'fixer-upper' in t or 'needs' in t: return {'compound': -0.55}
            else: return {'compound': 0.35}
    analyzer = SimpleSentimentAnalyzer()

# Apply NLP algorithm to score the positivity/negativity of the description
df['Text_Appeal_Score'] = df['Description'].apply(lambda x:
analyzer.polarity_scores(x)['compound'])

print("--- Step 2: Data After Text Mining Feature Engineering ---")
display(df[['Description', 'Text_Appeal_Score', 'Price']].head())
# 2. Predictive Modeling: Multiple Linear Regression
features = ['SqFt', 'Bedrooms', 'Age', 'Text_Appeal_Score']
X = df[features]
y = df['Price']

# Split Data (80% Train, 20% Test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
# Train the Model
model = LinearRegression()
model.fit(X_train, y_train)

# 3. Model Evaluation
y_pred = model.predict(X_test)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("\n--- Step 3: Predictive Model Performance (On Hidden Test Data) ---")
print(f"Test R-Squared (Accuracy): {r2:.4f}")
print(f"Test RMSE (Error margin):   ${rmse:,.2f}")

print("\n--- Step 4: Feature Importance (Coefficients) ---")
coef_df = pd.DataFrame({'Feature': features, 'Financial Impact (Coefficient)': model.coef_})
coef_df['Financial Impact (Coefficient)'] = coef_df['Financial Impact (Coefficient)'].apply(lambda
x: f"${x:,.2f}")
display(coef_df)
```

Part 4: Writing the Professional Report
Your deliverable is a formal business report. You are writing this for the executive board of UrbanTrend Real Estate Analytics.

Open a blank Word document and format your report using the exact headings below. Write in professional paragraph format. Do not simply list bulleted answers to the prompt questions.

- **Executive Summary** (1 paragraph)
  - Summarize the purpose of this project. Mention what types of data were used (structured and unstructured) and what the ultimate goal of the predictive algorithm was.
- **Exploratory Data Analysis** (1-2 paragraphs)
  - Review the two charts generated in Part 2.
  - Describe the general relationship between Square Footage and Price.
  - Looking at the boxplot, does having more bedrooms always guarantee a higher median price? What happens to the price variance (the size of the boxes) as you move from 2 bedrooms to 5 bedrooms?

- **Text Mining Methodology & Value** (1-2 paragraphs)
  - Explain what was done in Step 2 of the code.
  - Explain to the executives how the unstructured text descriptions were converted into a usable number (Text_Appeal_Score).
  - Look at the "Data After Text Mining" table. Notice how highly enthusiastic descriptions got a high score (near 0.8+) and "fixer-upper" descriptions got a negative score. Why is extracting this metric critical before building a pricing model?

- **Predictive Analytics Results** (2 paragraphs)
  - Review the output from Step 3 and Step 4 of the Jupyter script.
  - **Accuracy:** State the Test R^2 value and explain what it means regarding the model's ability to predict real-world data it hasn't seen before. Mention the RMSE (average error margin in dollars).
  - **Business Drivers:** Look at the "Feature Importance (Coefficients)" table. According to your

- model, exactly how much financial value does adding one extra Bedroom bring to a house? Conversely, how much financial value is lost for every 1 year of the home's Age?
- **The Value of Text:** Look at the coefficient for Text_Appeal_Score. Because VADER scores range from -1.0 to +1.0, a jump from a neutral description (0.0) to a highly positive description (+1.0) represents an increase of 1.0 unit. Based on your coefficient, how much is an excellent, highly positive written description worth in dollars?

- **Strategic Recommendations** (2 bullet points)
  - Based on your findings, give two actionable recommendations to real estate agents selling homes (e.g., Should they spend time crafting compelling property descriptions? Should they focus on marketing large square footage or older homes?).

Part 5: Final Submission Instructions

1. **Save your Written Report:** Save your Word document as a PDF (LastName_Assignment2_Report.pdf).
2. **Export your JupyterLab Notebook:**
   - Make sure all code cells have been executed sequentially and outputs (tables/charts) are displayed.
   - Go to **File** > **Print...** in the JupyterLab top menu bar.
   - Select **Save as PDF** as the destination printer and save as (LastName_Assignment2_JupyterLab.pdf).
3. **Combine as an Appendix:**
   - Merge the two PDFs into a single file. Ensure your Written Report is the first section, and the JupyterLab output is attached at the end as an Appendix.
   - Name the final merged file YourLastName_DATA300_Assignment2.pdf and submit it to the course portal.

Grading Expectations

Successful completion of this assignment requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Assignment 2 Predictive Data Mining Report Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

Week 6

Learning Materials

Read

Wall Street Prep. (n.d.). *Financial modeling guide | Excel training tutorial*. https://www.wallstreetprep.com/knowledge/financial-modeling/

Workday. (n.d.). *Financial forecasting in Excel made easy*. https://blog.workday.com/en-us/financial-forecasting-in-excel-made-easy.html

Software / Platforms

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Watch

Excel forecasting: Data analysis & predictive modeling tutorial. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=D54hg-_BtuQ

Intro to business intelligence | Part 1. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=dn97ux9exbY

Lab 6 Advanced Spreadsheet Simulation CLO 4, 5

Overview

This lab introduces spreadsheet simulation techniques for evaluating business decisions under uncertainty. By the end of this lab, students will be able to develop spreadsheet-based simulation models to evaluate alternative business scenarios (CLO 4) while assessing how analytical models support organizational decision-making (CLO 5). Spreadsheet simulations are widely used in finance, operations, project management, and risk analysis to improve strategic planning. The experience gained in this lab prepares students to integrate multiple analytical techniques within comprehensive business decision models.

Objective: This hands-on laboratory session engages you in the practical application of spreadsheet-based simulations. You will configure dynamic decision support models in Microsoft Excel to simulate varied business scenarios under uncertainty.

The Business Scenario: EcoDrive Motors: You are a financial analyst for **EcoDrive Motors**, a company planning to launch a new, affordable Electric Vehicle (EV). The profitability of this launch is highly uncertain due to fluctuating battery material costs and unpredictable consumer demand.

Furthermore, executive leadership is debating the supply chain. You can source batteries from a supplier known for severe labor violations (cheap) or an ethically certified supplier (an additional premium of $500 per vehicle).

The Task: Build an Excel simulation model to project financial outcomes, assess the risk of a financial loss, and evaluate the ethical implications of the supply chain decision.

Part 1: Building the Base Deterministic Model

Open a blank Microsoft Excel workbook. Save it immediately as YourLastName_DATA300_Lab6.xlsx.

Step 1: Setup the Assumptions

1. Rename "Sheet1" to Base Model.

2. Enter the following labels in column A (A1 to A7):
   - A1: **Scenario Assumptions**
   - A2: Expected Unit Sales (Demand)
   - A3: EV Sale Price (Per Unit)
   - A4: Base Manufacturing Cost (Per Unit)
   - A5: Ethical Sourcing Premium (Per Unit)
   - A6: Fixed Overhead Costs

3. Enter the following values in column B (B2 to B6):
   - B2: 15000
   - B3: 35000
   - B4: 28000

- B5: 500
- B6: 50000000 *(That is 50 million)*
- *Format B3, B4, B5, and B6 as Currency ($).*

Step 2: Setup the Output Metrics

1. Enter the following labels in column D (D1 to D4):

- D1: **Financial Projections**
- D2: Total Revenue
- D3: Total Variable Costs
- D4: Net Profit

1. Enter the formulas in column E:

- In E2, type: =B2*B3 *(Demand * Sale Price)*
- In E3, type: =B2*(B4+B5) *(Demand * (Base Cost + Ethical Premium))*
- In E4, type: =E2-E3-B6 *(Revenue - Variable Costs - Fixed Overhead)*
- *Format column E as Currency ($).*
  Checkpoint: Your Net Profit in cell E4 should be $47,500,000.

Part 2: What-If Analysis (Data Table)

Because demand and manufacturing costs are uncertain, you will use Excel's **Data Table** feature to simulate 25 different future scenarios instantly.

1. Create a new worksheet and rename it to What-If Analysis.
2. In cell **B2**, link to your Net Profit calculation by typing: ='Base Model'!E4
3. **Setup the Rows (Varying Demand):** In cells A3 down to A7, enter five different demand scenarios:
   5000, 10000, 15000, 20000, 25000
4. **Setup the Columns (Varying Base Cost):** In cells C2 to G2, enter five different manufacturing cost scenarios:
   24000, 26000, 28000, 30000, 32000
5. **Create the Data Table:**
   - Highlight the entire table range from **A2 to G7**.
   - Go to the Excel Ribbon: **Data > What-If Analysis > Data Table...**
   - In the *Row input cell* box, click on cell **B4** in your Base Model sheet (the Base Manufacturing Cost).
   - In the *Column input cell* box, click on cell **B2** in your Base Model sheet (the Expected Demand).
   - Click **OK**.

Highlight the newly populated cells (C3:G7) and format them as Currency. Use **Conditional Formatting** (Color Scales: Green-Yellow-Red) so profitable scenarios are green and negative scenarios are red.

Part 3: Dynamic Risk Simulation (Monte Carlo)

We will now simulate 50 "parallel universes" where demand fluctuates randomly, allowing us to assess the mathematical risk of a financial loss.

1. Create a new worksheet and rename it to Risk Simulation.
2. Set up headers in Row 1:
   - A1: Trial Number
   - B1: Simulated Demand
   - C1: Simulated Net Profit

In column A (A2 to A51), type the numbers 1 through 50.

In cell B2, use Excel's random generator to simulate demand between 8,000 and 22,000 units. Type:
=RANDBETWEEN(8000, 22000)

In cell C2, calculate the profit for this specific random demand using the fixed variables from your Base Model. Type:
=(B2*'Base Model'!$B$3) - (B2*('Base Model'!$B$4+'Base Model'!$B$5)) - 'Base Model'!$B$6

Drag the formulas in B2 and C2 all the way down to row 51. *(Note: The numbers will change every time you edit the sheet. This is the simulation working!)*

Simulation Summary: In cells E1 to E4, create a summary box:
- E1: Average Expected Profit: | F1: =AVERAGE(C2:C51)
- E2: Minimum Profit (Worst Case): | F2: =MIN(C2:C51)
- E3: Maximum Profit (Best Case): | F3: =MAX(C2:C51)
- E4: Probability of Loss: | F4: =COUNTIF(C2:C51, "<0")/50 *(Format F4 as a Percentage)*

Part 4: Applied Analytics Follow-up Questions
Based on the Excel simulation you just built, answer the following questions in a separate Word document.

Predictive Modeling & Future Outcomes (CLO 4):
1. Look at your What-If Analysis Data Table. If the Base Manufacturing Cost spikes to $32,000 due to supply chain shortages, what is the absolute minimum number of vehicles (Demand) EcoDrive Motors must sell to avoid a financial loss?
2. Look at your Risk Simulation sheet. *(Press F9 on your keyboard a few times to recalculate the random numbers and observe the changes).* Based on the "Probability of Loss" metric, how risky is this vehicle launch? What is the average expected profit across your simulated trials?

Ethical & Societal Contexts (CLO 5):
3. **The Ethical Dilemma:** Go back to your Base Model sheet. Change the **Ethical Sourcing Premium** (B5) from $500 to $0. This represents sourcing batteries from a supplier with known, severe labor violations.
- How much does your Net Profit increase when you drop the ethical supplier?
- As an analytics manager, write a brief, 3-4 sentence recommendation to the executive board arguing *why* they should keep the $500 ethical premium, despite the data showing it reduces immediate net profit. Consider long-term brand reputation, societal impact, and corporate responsibility.

Societal Impact of BI Tools: How does building a spreadsheet simulation like this help a company like EcoDrive Motors responsibly bring an affordable EV to the market? Why is it socially and financially dangerous for a company to make pricing decisions based on a single "best-case scenario" rather than simulating uncertainty?

Part 5: Final Submission Instructions
1. Ensure your Excel file is saved as YourLastName_DATA300_Lab6.xlsx. Make sure your Ethical Sourcing Premium is set back to $500 before saving.
2. Ensure your Word document containing the answers to the 4 questions is saved as a PDF (YourLastName_Lab6_Answers.pdf).
3. Submit **both** the Excel file (.xlsx) and the PDF document to the course portal.

Grading Expectations

Successful completion of this lab requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Lab 6 Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

Week 7

Learning Materials

Read

Tableau. (n.d.). *Developing a business intelligence strategy*. https://www.tableau.com/business-intelligence/business-intelligence-strategy

ThoughtSpot. (n.d.). *How to develop a successful business intelligence strategy*. https://www.thoughtspot.com/data-trends/business-intelligence/business-intelligence-strategy

Software / Platforms

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Watch

Implementing a business intelligence strategy in your organization (15 minutes). (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=m1zTr7l8cwU

Lab 7: BI Case Study Implementation CLO 4, 5

Overview

This lab integrates business intelligence concepts through the analysis of a comprehensive organizational case study. By the end of this lab, students will be able to evaluate the effectiveness of business intelligence tools in solving organizational challenges (CLO 5) while applying predictive modeling techniques to support business decisions (CLO 4). Business intelligence professionals routinely integrate multiple analytical tools to transform organizational data into meaningful recommendations for stakeholders. This lab provides direct preparation for the comprehensive capstone assignment by combining the analytical competencies developed throughout the course.

Objective: This applied laboratory session engages you in a practical application of business intelligence (BI) software. Utilize Microsoft Excel's advanced predictive tools to evaluate a business case study, drawing connections across descriptive and predictive metrics. Synthesize these models to generate actionable managerial reports that drive ethical and effective strategic decisions.

The Business Scenario: EcoDrive Motors (Phase 2)

Following your initial risk simulation in Lab 6, the executive board at **EcoDrive Motors** has a new concern. Global shortages of lithium and cobalt have caused the base cost of EV batteries to fluctuate wildly over the last three years.

Before committing to the $500 Ethical Sourcing Premium for your new vehicle launch next year, the Chief Financial Officer (CFO) needs to know where battery costs are heading.

The Task:
1. Use Excel's BI forecasting tools to build a predictive time series model of future battery costs.
2. Build a Case Study Dashboard comparing three different sourcing strategies under these new projected costs.
3. Generate a managerial report evaluating the financial and ethical impacts of these strategies.

Part 1: Time Series Forecasting in Excel (CLO 4)
Open a blank Microsoft Excel workbook. Save it as YourLastName_DATA300_Lab7.xlsx.

Step 1: Input Historical Data
1. Rename "Sheet1" to Historical Data.
2. Set up your headers in cells A1 and B1:
- A1: Quarter
- B1: Battery Cost per EV ($)
3. Enter the following historical data into A2:B13 (representing the last 3 years):
- A2: Q1-2023 | B2: 7500
- A3: Q2-2023 | B3: 7650
- A4: Q3-2023 | B4: 7900
- A5: Q4-2023 | B5: 8100
- A6: Q1-2024 | B6: 8000
- A7: Q2-2024 | B7: 8350
- A8: Q3-2024 | B8: 8600
- A9: Q4-2024 | B9: 8900
- A10: Q1-2025| B10: 8800
- A11: Q2-2025| B11: 9200
- A12: Q3-2025| B12: 9550
- *(Format Column B as Currency)*

Step 2: Generate the Predictive Model
Excel has a built-in Exponential Smoothing (ETS) algorithm that detects trends and seasonality to project future outcomes.
1. Highlight your entire dataset (A1 to B13).
2. Go to the Excel Ribbon: Click the **Data** tab.
3. In the "Forecast" group, click **Forecast Sheet**.
4. A window will pop up showing a preview of your time series forecast.
- Ensure "Forecast End" is set to point to **Q4-2026** (it may show as a date or row number depending on your Excel version; ensure it forecasts 4 steps ahead).

- Click **Create**.
- Excel will automatically generate a new worksheet containing the predictive graph, the forecasted values, and the 95% Confidence Intervals (Upper and Lower bounds) for the year 2026.
- Rename this new worksheet to Predictive Forecast.

Part 2: BI Case Study Dashboard
Apply the worst-case scenario from your forecast to evaluate three distinct business strategies.
1. Create a new worksheet and rename it to Case Study Dashboard.
2. **Setup the Constants (A1:B4):**
- A1: Expected EV Demand | B1: 15000

- A2: EV Sale Price | B2: 35000
- A3: Non-Battery Manufacturing Cost | B3: 18000
- A4: Fixed Overhead Costs | B4: 50000000

3. Pull in the Predictive Data (A6:B6):
   - A6: Projected Battery Cost (Upper Confidence Bound for Q4-2026)
   - B6: Type = and then click over to your Predictive Forecast sheet and select the cell containing the **Upper Confidence Bound** for the very last forecasted period (Q4-2026). Press Enter. *(This represents the worst-case predicted material cost).*

Step 3: Evaluate Three Sourcing Strategies
Set up a comparison table starting in row 8:
- A8: Strategy
- B8: 1. Profit-Maximized (No Ethics)
- C8: 2. Hybrid Sourcing (50% Ethical)
- D8: 3. Fully Ethical (100% Certified)

Row 9: Ethical Premium Cost per EV
- A9: Ethical Premium
- B9: 0
- C9: 250 *(Half of the $500 premium)*
- D9: 500

Row 10: Total Projected Variable Cost per EV
- A10: Total Variable Cost
- B10: =$B$3+$B$6+B9
- C10: =$B$3+$B$6+C9
- D10: =$B$3+$B$6+D9

Row 11: Projected Net Profit
- A11: Projected Net Profit
- B11: =($B$1*$B$2) - ($B$1*B10) - $B$4
- C11: =($B$1*$B$2) - ($B$1*C10) - $B$4
- D11: =($B$1*$B$2) - ($B$1*D10) - $B$4

Format rows 9, 10, and 11 as Currency ($).

Part 3: Managerial Report & Follow-up Questions
Based on the predictive models and BI dashboard you just built, write a formal Managerial Report in a separate Word document answering the following questions.

Section 1: Predictive Modeling Analysis (CLO 4)
1. **Trend Analysis:** Look at the chart and data generated in your Predictive Forecast sheet. Describe the historical trend of battery costs. What is the baseline forecasted battery cost for Q4-2026?
2. **Confidence Intervals:** Excel generated a "Lower Confidence Bound" and an "Upper Confidence Bound." Explain in your own words what these bounds represent in predictive modeling. Why did we choose to use the *Upper* bound in our Case Study Dashboard rather than the baseline forecast?

Section 2: Ethical Impact & BI Evaluation (CLO 5)

3. **Financial Impact of Ethics:** Look at your Case Study Dashboard. Assuming the worst-case projected battery costs for Q4-2026 become reality, what is the projected Net Profit for Strategy 1 (No Ethics) versus Strategy 3 (Fully Ethical)?
4. **Societal Context & Decision Making:** EcoDrive Motors markets itself as a "Green and Responsible" company. However, the data shows that 100% ethical sourcing combined with rising forecasted material costs will significantly reduce profit margins. As an analytics consultant, write a 1-2 paragraph executive recommendation detailing which of the three strategies the company should adopt. Justify your choice by balancing the mathematical data (profits/costs) against societal impacts (brand reputation, human rights, corporate social responsibility).
5. **The Value of BI:** How does combining a predictive Time Series forecast with a multi-strategy BI dashboard lead to better corporate decision-making than simply looking at last year's costs?

Part 4: Final Submission Instructions

1. Ensure your Excel file is completely filled out and saved as YourLastName_DATA300_Lab7.xlsx.
2. Ensure your Word document containing the Managerial Report is professionally formatted and saved as a PDF (YourLastName_Lab7_Report.pdf).
3. Submit **both** the Excel file (.xlsx) and the PDF report document to the course portal.

Grading Expectations

Successful completion of this lab requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Lab 7 Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

Assignment 3 Spreadsheet Simulation and Decision Model CLO 3, 4, 5

Overview

This assignment applies spreadsheet simulation techniques to evaluate alternative business strategies using quantitative decision models. By the end of this assignment, students will be able to apply computational problem-solving techniques to business scenarios (CLO 3), develop predictive simulation models that project future organizational outcomes (CLO 4), and evaluate the impact of analytical results on strategic decision-making (CLO 5). Spreadsheet simulation remains one of the most widely used decision-support tools across finance, operations, and executive planning. This assignment strengthens the analytical reasoning and modeling skills required for the comprehensive capstone project.

Objective: This assignment requires the development of a spreadsheet-based predictive simulation model. Build upon simulated data mining insights to construct a dynamic forecasting tool in Microsoft Excel that models potential future business outcomes under uncertainty. You will then export this simulated data into Tableau to visually evaluate the outcomes and formulate a rigorous organizational strategy.

The Business Scenario: You are an analytics planner for "SmartHome Tech," preparing for the launch of a new AI-powered thermostat. Based on previous data mining efforts, your team has discovered two key insights:

1. Baseline demand is roughly 1,000 units per month, but historical "market volatility" causes this to fluctuate randomly by +/- 150 units.
2. The thermostat yields a strict $100 profit margin per unit.
   You need to simulate 12 months of future sales to decide between two marketing strategies:
   - **Strategy A (Standard):** Fixed marketing cost of $10,000/month. Yields standard baseline demand.
   - **Strategy B (Aggressive):** Fixed marketing cost of $25,000/month. Multiplies demand by 1.4x.

Part 1: Spreadsheet Simulation in Microsoft Excel
Open a blank Microsoft Excel workbook. Save it immediately as YourLastName_DATA300_Assignment3.xlsx.

Step 1: Set Up the Simulation Headers
In Row 1, enter the following headers from columns A through G:
- A1: Month
- B1: Baseline_Demand
- C1: Market_Volatility
- D1: Standard_Demand
- E1: Aggressive_Demand
- F1: Standard_Profit
- G1: Aggressive_Profit

Step 2: Build the Simulation Formulas
1. **Month & Baseline:** In cells A2 down to A13, type the numbers 1 through 12. In cells B2 down to B13, type 1000 (the baseline demand).
2. **Market Volatility (Randomness):** In cell C2, type =RANDBETWEEN(-150, 150). Drag this formula down to C13 to simulate random monthly market swings.
3. **Standard Demand:** In cell D2, type =B2+C2. Drag down to D13.
4. **Aggressive Demand:** In cell E2, type =ROUND((B2+C2)*1.4, 0). Drag down to E13.
5. **Standard Profit:** In cell F2, type =(D2*100)-10000 *(Demand * $100 margin - $10k marketing)*. Drag down to F13.
6. **Aggressive Profit:** In cell G2, type =(E2*100)-25000 *(Demand * $100 margin - $25k marketing)*. Drag down to G13.

Step 3: Lock the Simulation & Export
Because RANDBETWEEN changes every time you click, we must "lock" the simulation before moving to Tableau.
1. Highlight your entire table (A1 to G13).
2. Press **Ctrl+C** (or Cmd+C) to Copy.
3. Without moving your cursor, right-click cell A1 and select **Paste Special > Values** (the icon with the "123" on a clipboard).
4. Your random numbers are now locked. Save your Excel file and close Excel.

Part 2: Decision Modeling in Tableau
Now, we will visualize our simulated future in Tableau to make a business decision.
- Open **Tableau Desktop**.
- On the left under "Connect", click **Microsoft Excel** and select your saved YourLastName_DATA300_Assignment3.xlsx file. Click the orange **Sheet 1** tab at the bottom to begin.

- **Set the Timeline:** Drag **Month** from the left pane into the **Columns** shelf.
- *Important:* Right-click the green 'Month' pill in the Columns shelf and select **Dimension**.
- **Plot the Strategies:** * Drag **Measure Values** (from the bottom left of the data pane) into the **Rows** shelf.
- Tableau will draw many lines. Look for the "Measure Values" card that appeared below the "Marks" card. Remove everything from this list *except* **SUM(Standard Profit)** and **SUM(Aggressive Profit)** by right-clicking and selecting "Remove".
- **Format the Visual:**
- In the **Marks** card, change the dropdown from "Automatic" to **Line**.
- Drag **Measure Names** from the data pane and drop it onto the **Color** square in the Marks card to give each strategy a different colored line.
- **Save the Visual:** Go to **Worksheet > Export > Image...**, save it as a PNG or JPEG file, and insert this image into a blank Word document.

Part 3: Professional Report & Follow-up Questions
In the Word document where you pasted your Tableau visual, write a brief executive report answering the following two questions.

1. Strategic Decision Making:
Look at your Tableau visualization comparing the two profit lines over the 12-month simulation. Overall, which marketing strategy (Standard or Aggressive) yields the highest and most consistent profit? Based on your specific simulation, were there any months where the Aggressive strategy performed *worse* financially than the Standard strategy due to its high fixed costs during a "down" volatility month?

2. The Value of Predictive Simulation:
In our Excel model, we utilized a random number generator (RANDBETWEEN) to simulate "Market Volatility" based on historical data mining. From a managerial perspective, why is injecting this random risk factor into a predictive forecasting model much safer for organizational planning than simply calculating profits assuming the company will sell exactly 1,000 units every single month?

Part 4: Final Submission Instructions
1. Ensure your short report answering the two questions includes your Tableau visual and is saved as a PDF (YourLastName_Assignment3_Report.pdf).
2. Submit **both** your locked Excel file (.xlsx) and your PDF report to the course portal.

Grading Expectations
Successful completion of this assignment requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Assignment 3 Spreadsheet Simulation and Decision Model Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

Week 8

Learning Materials

Read

Review all the materials in preparation for the SM Reflection and capstone project

Software / Platforms

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Assignment 4 Final Comprehensive Assignment (Capstone) CLO 4, 5

Overview

This comprehensive assignment integrates the analytical methods, business intelligence tools, and predictive modeling techniques developed throughout the course into a complete organizational case study. By the end of this assignment, students will be able to develop predictive models that support strategic decision-making (CLO 4) while evaluating the organizational impact of business intelligence solutions and communicating evidence-based managerial recommendations (CLO 5). These competencies reflect the complete analytical workflow performed by business intelligence professionals who transform organizational data into strategic business value. Successful completion of this capstone demonstrates the ability to integrate descriptive, predictive, and prescriptive analytics into comprehensive, data-driven business solutions.

Objective: This culminating integrative project synthesizes business intelligence concepts, predictive modeling, and data visualization into a comprehensive managerial report. You will evaluate a complex case study by connecting Python (JupyterLab Lab), Microsoft Excel, and Tableau. Your goal is to formulate data-driven strategic recommendations that balance financial viability with ethical corporate contexts, presenting your findings clearly to non-technical stakeholders.

The Business Scenario: You are the Lead Business Intelligence Analyst for **"EcoVate,"** an electronics company launching a new smart-home energy manager.

Before committing to a 5-year manufacturing contract, the executive team needs to answer two questions:

1. **Demand Forecasting:** Based on recent unstructured focus group feedback, how well will this product sell?
2. **Ethical Supply Chain:** Should EcoVate use a standard overseas manufacturer (cheaper, but with known carbon emission violations) or a "Fair-Trade Green Certified" manufacturer (more expensive, but ethically sound)?

Task: Analyze focus group text in JupyterLab Lab to generate a "Demand Multiplier." You will apply this multiplier to a 5-year financial simulation in Excel. Finally, you will export the financial data to Tableau to visualize the strategic choice for the executive board.

Part 1: Text Mining in JupyterLab Lab (Sentiment to Strategy)

First, we must analyze the raw, unstructured text from the focus groups to determine a mathematical adjustment for our sales forecast.

- Open your web browser and go to https://jupyter.org/try-jupyter/lab/.
- Once the workspace loads, click on **Python 3 (ipykernel)** or **Pyodide** under the **Notebook** section in the Launcher tab to create a new notebook.
- Rename your notebook file:
  - Go to the top menu bar and click **File** > **Rename Notebook...** (or right-click Untitled.ipynb in the left file browser tab).
  - Rename the notebook to: YourLastName_DATA300_Capstone.ipynb.

- Click inside the first empty code cell, copy and paste the Python code below, and press **Shift + Enter** (or click the **Play** ▶ button):

```
# Install required libraries in the active Jupyter kernel
%pip install -q pandas nltk

import pandas as pd
import nltk

# 1. Initialize NLP Lexicon with browser sandbox compatibility
try:
    nltk.download('vader_lexicon', quiet=True)
    from nltk.sentiment.vader import SentimentIntensityAnalyzer
    analyzer = SentimentIntensityAnalyzer()
except Exception:
    # WebAssembly/Browser sandbox fallback if socket requests are restricted
    class SimpleSentimentAnalyzer:
        def polarity_scores(self, text):
            t = text.lower()
            if 'love' in t or 'incredible' in t or 'brilliant' in t:
                return {'compound': 0.85}
            elif 'okay' in t or 'confusing' in t:
                return {'compound': 0.20}
            else:
                return {'compound': 0.50}
    analyzer = SimpleSentimentAnalyzer()

# 2. Unstructured Focus Group Feedback
reviews = [
    "I absolutely love the energy-saving features. It looks sleek and works perfectly.",
    "Good product, but the mobile app is a little bit confusing to set up.",
    "Incredible! I saved $40 on my electric bill the first month. Highly recommend.",
    "It is an okay device. Kind of expensive for what it actually does.",
    "Brilliant integration with my other smart devices. A must-buy for green homes."

]

df = pd.DataFrame({'Review_Text': reviews})

# 3. Apply Sentiment Analysis
df['Sentiment_Score'] = df['Review_Text'].apply(lambda x:
analyzer.polarity_scores(x)['compound'])

# 4. Convert Average Sentiment into a Business 'Demand Multiplier'
# Baseline demand is 1.0. We add up to 20% (0.2) based on positive sentiment.
avg_sentiment = df['Sentiment_Score'].mean()
demand_multiplier = 1.0 + (avg_sentiment * 0.20)

print("--- Step 1: Sentiment Analysis Results ---")
display(df)
print(f"\nAverage Sentiment Score: {avg_sentiment:.3f}")
```

```
print(f"--> YOUR CALCULATED DEMAND MULTIPLIER: {demand_multiplier:.3f} <--")
print("(Write this multiplier down for Part 2!)")
```

Exporting Your JupyterLab Output

- Ensure all outputs, tables, and calculated values are clearly displayed.
- Go to **File** > **Print...** in the JupyterLab top menu bar.
- Select **Save as PDF** as the destination printer and save the file as LastName_JupyterLabLab.pdf (or LastName_JupyterLab.pdf as instructed).
- **Note down your calculated Demand Multiplier** to use in your Excel financial modeling section (Part 2).

Part 2: Financial Simulation in Microsoft Excel

Apply your NLP insights to a financial model to compare the standard vs. ethical supply chains.

1. Open a blank Microsoft Excel workbook and save it as YourLastName_DATA300_Capstone.xlsx.
2. **Set up the Headers:** In Row 1, enter the following from A1 to F1:

- A1: Year
- B1: Base_Demand
- C1: Colab_Multiplier
- D1: Adjusted_Demand
- E1: Standard_Profit
- F1: Ethical_Profit

Enter the Baseline Data (Years and Base Demand):

- Cells A2 to A6: Type the numbers 1, 2, 3, 4, 5.
- Cells B2 to B6: Type the following base demand forecasts respectively: 10000, 15000, 22000, 30000, 45000.
- **Enter the Multiplier:** In cells C2 down to C6, type the exact **Demand Multiplier** number you generated in JupyterLab Lab (e.g., 1.121).
- **Calculate Adjusted Demand:** In cell D2, type =ROUND(B2*C2, 0). Drag this formula down to D6.
- **Calculate Profit Scenarios:** * *Assumptions:* The product sells for $200. Fixed costs are $500,000/year. The Standard supplier costs $120 to build (Margin: $80). The Ethical supplier costs $140 to build (Margin: $60).
- **Standard Profit:** In cell E2, type =(D2*80)-500000. Drag down to E6.
- **Ethical Profit:** In cell F2, type =(D2*60)-500000. Drag down to F6.
- Save the Excel file and close Excel.

Part 3: Executive Visualization in Tableau

Export your financial simulation into Tableau to create a visual comparison for the executive board.

1. Open **Tableau Desktop**. Under "Connect", click **Microsoft Excel** and open your saved .xlsx Capstone file. Click **Sheet 1**.
2. **Setup the Timeline:** Drag **Year** to the **Columns** shelf. Right-click the green pill and change it to **Dimension**.
3. **Plot the Profits:** Drag **Measure Values** to the **Rows** shelf.
4. Filter the "Measure Values" card (below the Marks card). Remove everything *except* **SUM(Standard Profit)** and **SUM(Ethical Profit)**.
5. **Format for the Executives:**

- In the **Marks** card, change the chart type to **Line** or **Bar** (whichever you feel is more impactful).
- Drag **Measure Names** to the **Color** square to differentiate the lines/bars.
- Double-click the Y-axis numbers to ensure the format is set to Currency.
- **Save the Visual:** Go to **Worksheet > Export > Image...** and save your chart. Insert this image into a blank Word document.

Part 4: Integrative Managerial Report
In the Word document containing your Tableau visual, write a professional executive summary consisting of a brief introduction and your answers to the following two integrative questions.

1. Strategic & Ethical Recommendation (CLO 5):
Review your Tableau visualization. As expected, the Ethical Supply Chain yields lower profits than the Standard Supply Chain across all 5 years. However, assume the focus group indicated that EcoVate's target demographic heavily values corporate carbon footprints**.**
As the Lead BI Analyst, write a recommendation to the executive board. Should they choose the Standard or Ethical supply chain? Justify your decision by balancing the projected financial data (how severe is the profit loss?) against the ethical context and long-term brand valuation.

2. The Value of Integrated BI Ecosystems:
In this project, you acted as a data translator across three platforms: NLP Text Mining (JupyterLab Lab) -> Spreadsheets (Excel) -> Visual Storytelling (Tableau). Explain why integrating these three specific steps provides a more robust and realistic strategic business plan than relying on a static spreadsheet alone.

Part 5: Final Submission Instructions
You have three deliverables for this final capstone:
1. Ensure your Word document containing your Tableau visual and report answers is saved as a PDF (YourLastName_Capstone_Report.pdf).
2. Attach your exported JupyterLab Lab PDF (LastName_JupyterLabLab.pdf) to the end of your report, OR submit it as a separate file.
3. Submit your completed Excel file (YourLastName_DATA300_Capstone.xlsx).

Submit all files to the final course portal.

Grading Expectations
Successful completion of this assignment requires accurate application of the statistical concepts, analytical techniques, and software tools presented in the module. Students are expected to complete all required tasks, submit the specified deliverables, and present their work in a clear and professional manner.

Students should review the **Assignment 4 Final Comprehensive Assignment (Capstone) Rubric** before beginning the activity to understand the evaluation criteria, performance expectations, and point distribution.

SM Activity 2 Reflection
CLO 1 - 5

Socratic Metric (SM) Assignment Instructions
This activity is designed to support student learning through verbal explanation, reflection, and revision. The process emphasizes developing conceptual understanding by articulating ideas clearly and responding to feedback.

Getting Started

- Review the Socratic Metric (SM) Student Guide to become familiar with system expectations and functionality.
- Copy the SM Assignment link provided by the instructor in the online course.
- Open the Socratic Metric Assignment Pop Up (indicated by the puzzle icon)
- Paste the link from the instructor in the pop up to begin

Assignment Process

- Read the Question Aloud
  - Speak clearly and at a natural pace.
  - Speech should be sufficiently clear for the system to capture at least 70% accuracy.
- Provide a Verbal Response
  - Explain the concept using your own words.
  - Respond using natural speech, as if explaining the idea in a real-world conversation.
- Use of Learning Resources
  - Reference textbooks or notes as needed.
  - Do not read directly from a script or AI-generated text.
- Review Feedback & Revising Responses
  - Read the system-generated feedback aloud after each submission.
  - Revise each response up to four times.

Submission Requirements

- Upon completion of the final question, select “Finish Assessment.”
- Remain on the page until the “Assignment Complete” screen appears to ensure successful submission.

Grading Criteria

The grading rubric for this assignment evaluates the overall quality of student responses, including the accuracy and relevance of information, effective use of course materials, and the depth of critical thinking demonstrated. View the Socratic Metric grading rubric, and become familiar with the expectations prior to beginning the assignment.

Course Rubrics

Activities Rubric

Week 1 Self-Assessment (Knowledge Quiz) (JupyterLab Lab and Tableau Readiness Check) Rubric

Criterion	Exemplary (100%)	Excellent (90%)	Proficient (75%)	Developing (60%)	Unsatisfactory (50%)	No submission or unacceptable submission (0%)
CLO 1 Describe the principles of descriptive, predictive, and prescriptive analytics to	Exceeds expectations describing the principles of descriptive, predictive, and prescriptive	Effectively describes the principles of descriptive, predictive, and prescriptive	Sufficiently describes the principles of descriptive, predictive, and prescriptive analytics to	Partially describes the principles of descriptive, predictive, and prescriptive	Does not describe the principles of descriptive, predictive, and prescriptive analytics to	No submission or unacceptable submission *0 points*

solve business data challenges. *20 Points*	analytics to solve business data challenges. *20 Points*	analytics to solve business data challenges. *18 points*	solve business data challenges. *15 points*	analytics to solve business data challenges. *12 points*	solve business data challenges. *10 points*	
CLO 2 Design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *20 Points*	Exceeds expectations designing and utilizing data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *20 Points*	Effectively designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *18 points*	Sufficiently designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *15 points*	Partially designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *12 points*	Does not design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *10 points*	No submission or unacceptable submission *0 points*
CLO 3 Apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *20 Points*	Exceeds expectations applying computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *20 Points*	Effectively applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *18 points*	Sufficiently applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *15 points*	Partially applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *12 points*	Does not apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *10 points*	No submission or unacceptable submission *0 points*
CLO 4 Develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *20 Points*	Exceeds expectations developing predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *20 Points*	Effectively develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *18 points*	Sufficiently develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *15 points*	Partially develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *12 points*	Does not develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *10 points*	No submission or unacceptable submission *0 points*
Scholarly & Professional Formatting: Follows applicable APA and/or discipline-specific formatting	**Exceeds expectations** following applicable APA and/or discipline-specific formatting requirements,	**Effectively** follows applicable APA and/or discipline-specific formatting requirements, including	**Sufficiently** follows applicable APA and/or discipline-specific formatting requirements, including	**Partially** follows applicable APA and/or discipline-specific formatting requirements, including	**Does not** follow applicable APA and/or discipline-specific formatting requirements, including accurate	No submission or unacceptable submission *0 points*

requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *9 points*	accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *7.5 points*	accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *6 points*	citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *5 points*	
Academic Writing and Mechanics Demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	Exceeds expectations in demonstrating academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	Effectively demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *9 points*	Sufficiently demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *7.5 points*	Partially demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *6 points*	Does not demonstrate academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *5 points*	No submission or unacceptable submission *0 points*

Socratic Metric Rubric

Criteria	Excellent (4)	Proficient (3)	Developing (2)	Needs Improvement (1)
Accuracy of Information	All responses are factually accurate and demonstrate deep understanding of the topic.	Most responses are accurate with minor errors that do not impact overall understanding.	Some responses contain errors or misconceptions affecting comprehension.	Many responses are inaccurate or show a lack of understanding.

Use of Course Material	Effectively integrates and applies course concepts to responses.	Uses course material correctly but may lack depth in application.	Limited reference to course concepts or misapplied information.	No connection to course material is evident.
Relevance to Prompt	Fully addresses the prompt with clear alignment to the discussion question.	Addresses the prompt but may lack depth or specificity.	Partially addresses the prompt but lacks clarity or relevance.	Does not address the prompt or response is off-topic.
Critical Thinking / Depth of Response	Demonstrates original thought, insightful analysis, and a strong ability to evaluate and synthesize information. Responses show clear reasoning and depth.	Shows solid reasoning and some analysis. May include basic evaluation of ideas but lacks deeper synthesis or originality.	Limited analysis or critical thought; responses tend to summarize rather than engage with the material.	Little to no evidence of analysis, reflection, or critical engagement. Responses are surface-level or purely descriptive.

Lab Rubrics

Lab 1 Rubric, Lab 2 Rubric, Lab 3 Rubric

Criterion	**Exemplary (100%)**	**Excellent (90%)**	**Proficient (75%)**	**Developing (60%)**	**Unsatisfactory (50%)**	**No submission or unacceptable submission (0%)**
CLO 1 Describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *25 points*	Exceeds expectations describing the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *25 points*	Effectively describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *22.5 points*	Sufficiently describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *18.75 points*	Partially describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *15 points*	Does not describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *12.5 points*	No submission or unacceptable submission *0 points*
CLO 2 Design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *25 points*	Exceeds expectations designing and utilizing data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *25 points*	Effectively designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *22.5 points*	Sufficiently designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *18.75 points*	Partially designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *15 points*	Does not design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *12.5 points*	No submission or unacceptable submission *0 points*

CLO 3 Apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *30 points*	Exceeds expectations applying computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *30 points*	Effectively applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *27 points*	Sufficiently applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *22.5 points*	Partially applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *18 points*	Does not apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *15 points*	No submission or unacceptable submission *0 points*
Scholarly & Professional Formatting: Follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Exceeds expectations** following applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Effectively** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *9 points*	**Sufficiently** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *7.5 points*	**Partially** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *6 points*	**Does not** follow applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *5 points*	No submission or unacceptable submission *0 points*
Academic Writing and Mechanics Demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling,	Exceeds expectations in demonstrating academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation,	Effectively demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation,	Sufficiently demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence	Partially demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation,	Does not demonstrate academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and	No submission or unacceptable submission *0 points*

punctuation, sentence structure, and discipline-appropriate language. *10 points*	sentence structure, and discipline-appropriate language. *10 points*	sentence structure, and discipline-appropriate language. *9 points*	structure, and discipline-appropriate language. *7.5 points*	sentence structure, and discipline-appropriate language. *6 points*	discipline-appropriate language. *5 points*	

Lab 4 Rubric

Criterion	**Exemplary (100%)**	**Excellent (90%)**	**Proficient (75%)**	**Developing (60%)**	**Unsatisfactory (50%)**	**No submission or unacceptable submission (0%)**
CLO 1 Describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *25 points*	Exceeds expectations describing the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *25 points*	Effectively describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *22.5 points*	Sufficiently describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *18.75 points*	Partially describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *15 points*	Does not describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *12.5 points*	No submission or unacceptable submission *0 points*
CLO 2 Design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *25 points*	Exceeds expectations designing and utilizing data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *25 points*	Effectively designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *22.5 points*	Sufficiently designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *18.75 points*	Partially designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *15 points*	Does not design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *12.5 points*	No submission or unacceptable submission *0 points*
CLO 5 Evaluate the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering	Exceeds expectations evaluating the impact of business intelligence tools within a comprehensive case study to generate managerial reports	Effectively evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering	Sufficiently evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering	Partially evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering	Does not evaluate the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering	No submission or unacceptable submission *0 points*

ethical and societal contexts. *30 points*	considering ethical and societal contexts. *30 points*	ethical and societal contexts. *27 points*	ethical and societal contexts. *22.5 points*	ethical and societal contexts. *18 points*	ethical and societal contexts. *15 points*	
Scholarly & Professional Formatting: Follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Exceeds expectations** following applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Effectively** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *9 points*	**Sufficiently** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *7.5 points*	**Partially** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *6 points*	**Does not** follow applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *5 points*	No submission or unacceptable submission *0 points*
Academic Writing and Mechanics Demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	Exceeds expectations in demonstrating academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	Effectively demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *9 points*	Sufficiently demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *7.5 points*	Partially demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *6 points*	Does not demonstrate academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *5 points*	No submission or unacceptable submission *0 points*

Lab 5 Rubric, Lab 6 Rubric, Lab 7 Rubric

Criterion	Exemplary (100%)	Excellent (90%)	Proficient (75%)	Developing (60%)	Unsatisfactory (50%)	No submission or unacceptable submission (0%)
CLO 4 Develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *40 points*	Exceeds expectations developing predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *40 points*	Effectively develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *36 points*	Sufficiently develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *30 points*	Partially develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *24 points*	Does not develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *20 points*	No submission or unacceptable submission *0 points*
CLO 5 Evaluate the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *40 points*	Exceeds expectations evaluating the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *40 points*	Effectively evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *36 points*	Sufficiently evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *30 points*	Partially evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *24 points*	Does not evaluate the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *20 points*	No submission or unacceptable submission *0 points*
Scholarly & Professional Formatting: Follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent	**Exceeds expectations** following applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of	**Effectively** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and	**Sufficiently** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and	**Partially** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and	**Does not** follow applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions.	No submission or unacceptable submission *0 points*

application of scholarly and professional conventions. *10 points*	scholarly and professional conventions. *10 points*	professional conventions. *9 points*	professional conventions. *7.5 points*	consistent application of scholarly and professional conventions. *6 points*	*5 points*	
Academic Writing and Mechanics Demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	Exceeds expectations in demonstrating academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	Effectively demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *9 points*	Sufficiently demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *7.5 points*	Partially demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *6 points*	Does not demonstrate academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *5 points*	No submission or unacceptable submission *0 points*

Assignment Rubrics

Assignment 1 Data Visualization and Descriptive Analytics Report Rubric

Criterion	**Exemplary (100%)**	**Excellent (90%)**	**Proficient (75%)**	**Developing (60%)**	**Unsatisfactory (50%)**	**No submission or unacceptable submission (0%)**
CLO 1 Describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *25 points*	Exceeds expectations describing the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *25 points*	Effectively describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *22.5 points*	Sufficiently describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *18.75 points*	Partially describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *15 points*	Does not describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *12.5 points*	No submission or unacceptable submission *0 points*
CLO 2 Design and utilize data visualization and modeling tools to	Exceeds expectations designing and utilizing data visualization and modeling	Effectively designs and utilize data visualization and modeling tools to	Sufficiently designs and utilize data visualization and modeling tools to	Partially designs and utilize data visualization and modeling tools to	Does not design and utilize data visualization and modeling tools to	No submission or unacceptable submission *0 points*

effectively communicate user-centered analytics for strategic decisions. *25 points*	tools to effectively communicate user-centered analytics for strategic decisions. *25 points*	effectively communicate user-centered analytics for strategic decisions. *22.5 points*	effectively communicate user-centered analytics for strategic decisions. *18.75 points*	effectively communicate user-centered analytics for strategic decisions. *15 points*	effectively communicate user-centered analytics for strategic decisions. *12.5 points*	
CLO 3 Apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *30 points*	Exceeds expectations applying computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *30 points*	Effectively applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *27 points*	Sufficiently applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *22.5 points*	Partially applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *18 points*	Does not apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *15 points*	No submission or unacceptable submission *0 points*
Scholarly & Professional Formatting: Follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Exceeds expectations** following applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Effectively** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *9 points*	**Sufficiently** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *7.5 points*	**Partially** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *6 points*	**Does not** follow applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *5 points*	No submission or unacceptable submission *0 points*
Academic Writing and Mechanics Demonstrates academic writing through	Exceeds expectations in demonstrating academic writing through clear,	Effectively demonstrates academic writing through clear, organized,	Sufficiently demonstrates academic writing through clear, organized, and	Partially demonstrates academic writing through clear, organized,	Does not demonstrate academic writing through clear, organized, and	No submission or unacceptable submission *0 points*

clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *9 points*	professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *7.5 points*	and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *6 points*	professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *5 points*	

Assignment 2 Predictive Data Mining Report Rubric

Criterion	**Exemplary (100%)**	**Excellent (90%)**	**Proficient (75%)**	**Developing (60%)**	**Unsatisfactory (50%)**	**No submission or unacceptable submission (0%)**
CLO 1 Describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *20 points*	Exceeds expectations describing the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *20 points*	Effectively describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *18 points*	Sufficiently describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *15 points*	Partially describes the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *12 points*	Does not describe the principles of descriptive, predictive, and prescriptive analytics to solve business data challenges. *10 points*	No submission or unacceptable submission *0 points*
CLO 2 Design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *20 points*	Exceeds expectations designing and utilizing data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *20 points*	Effectively designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *18 points*	Sufficiently designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *15 points*	Partially designs and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *12 points*	Does not design and utilize data visualization and modeling tools to effectively communicate user-centered analytics for strategic decisions. *10 points*	No submission or unacceptable submission *0 points*
CLO 3 Apply computational problem-solving to perform data and text mining, discovering	Exceeds expectations applying computational problem-solving to perform data and text mining,	Effectively applies computational problem-solving to perform data and text mining, discovering	Sufficiently applies computational problem-solving to perform data and text mining, discovering	Partially applies computational problem-solving to perform data and text mining, discovering	Does not apply computational problem-solving to perform data and text mining, discovering relationships in both structured	No submission or unacceptable submission *0 points*

relationships in both structured and unstructured data. *20 points*	discovering relationships in both structured and unstructured data. *20 points*	relationships in both structured and unstructured data. *18 points*	relationships in both structured and unstructured data. *15 points*	relationships in both structured and unstructured data. *12 points*	and unstructured data. *10 points*	
CLO 4 Develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *20 points*	Exceeds expectations developing predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *20 points*	Effectively develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *18 points*	Sufficiently develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *15 points*	Partially develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *12 points*	Does not develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *10 points*	No submission or unacceptable submission *0 points*
Scholarly & Professional Formatting: Follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Exceeds expectations** following applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Effectively** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *9 points*	**Sufficiently** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *7.5 points*	**Partially** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *6 points*	**Does not** follow applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *5 points*	No submission or unacceptable submission *0 points*
Academic Writing and Mechanics Demonstrates academic writing through clear, organized, and professional	Exceeds expectations in demonstrating academic writing through clear, organized, and professional language,	Effectively demonstrates academic writing through clear, organized, and professional language,	Sufficiently demonstrates academic writing through clear, organized, and professional language, using	Partially demonstrates academic writing through clear, organized, and professional language,	Does not demonstrate academic writing through clear, organized, and professional language, using appropriate	No submission or unacceptable submission *0 points*

language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *9 points*	appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *7.5 points*	using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *6 points*	grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *5 points*	

Assignment 3 Spreadsheet Simulation and Decision Model Rubric

Criterion	**Exemplary (100%)**	**Excellent (90%)**	**Proficient (75%)**	**Developing (60%)**	**Unsatisfactory (50%)**	**No submission or unacceptable submission (0%)**
CLO 3 Apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *25 points*	Exceeds expectations applying computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *25 points*	Effectively applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *22.5 points*	Sufficiently applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *18.75 points*	Partially applies computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *15 points*	Does not apply computational problem-solving to perform data and text mining, discovering relationships in both structured and unstructured data. *12.5 points*	No submission or unacceptable submission *0 points*
CLO 4 Develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *25 points*	Exceeds expectations developing predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *25 points*	Effectively develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *22.5 points*	Sufficiently develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *18.75 points*	Partially develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *15 points*	Does not develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *12.5 points*	No submission or unacceptable submission *0 points*
CLO 5 Evaluate the impact of business intelligence	Exceeds expectations evaluating the impact of business	Effectively evaluates the impact of business intelligence	Sufficiently evaluates the impact of business intelligence	Partially evaluates the impact of business intelligence	Does not evaluate the impact of business intelligence	No submission or unacceptable submission *0 points*

tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *30 points*	intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *30 points*	tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *27 points*	tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *22.5 points*	tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *18 points*	tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *15 points*	
Scholarly & Professional Formatting: Follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Exceeds expectations** following applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	**Effectively** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *9 points*	**Sufficiently** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *7.5 points*	**Partially** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *6 points*	**Does not** follow applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *5 points*	No submission or unacceptable submission *0 points*
Academic Writing and Mechanics Demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-	Exceeds expectations in demonstrating academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-	Effectively demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language.	Sufficiently demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language.	Partially demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language.	Does not demonstrate academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language.	No submission or unacceptable submission *0 points*

appropriate language. *10 points*	appropriate language. *10 points*	*9 points*	*7.5 points*	*6 points*	*5 points*	

Assignment 4 Final Comprehensive Assignment (Capstone)

Criterion	**Exemplary (100%)**	**Excellent (90%)**	**Proficient (75%)**	**Developing (60%)**	**Unsatisfactory (50%)**	**No submission or unacceptable submission (0%)**
CLO 4 Develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *40 points*	Exceeds expectations developing predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *40 points*	Effectively develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *36 points*	Sufficiently develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *30 points*	Partially develops predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *24 points*	Does not develop predictive models using time series forecasting, linear regression, and spreadsheet simulations to project future outcomes. *20 points*	No submission or unacceptable submission *0 points*
CLO 5 Evaluate the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *40 points*	Exceeds expectations evaluating the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *40 points*	Effectively evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *36 points*	Sufficiently evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *30 points*	Partially evaluates the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *24 points*	Does not evaluate the impact of business intelligence tools within a comprehensive case study to generate managerial reports considering ethical and societal contexts. *20 points*	No submission or unacceptable submission *0 points*
Scholarly & Professional Formatting: Follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references	**Exceeds expectations** following applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required,	**Effectively** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate	**Sufficiently** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate	**Partially** follows applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when	**Does not** follow applicable APA and/or discipline-specific formatting requirements, including accurate citations and references when required, appropriate formatting of the	No submission or unacceptable submission *0 points*

when required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *10 points*	formatting of the deliverable, and consistent application of scholarly and professional conventions. *9 points*	formatting of the deliverable, and consistent application of scholarly and professional conventions. *7.5 points*	required, appropriate formatting of the deliverable, and consistent application of scholarly and professional conventions. *6 points*	deliverable, and consistent application of scholarly and professional conventions. *5 points*	
Academic Writing and Mechanics Demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	Exceeds expectations in demonstrating academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *10 points*	Effectively demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *9 points*	Sufficiently demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *7.5 points*	Partially demonstrates academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *6 points*	Does not demonstrate academic writing through clear, organized, and professional language, using appropriate grammar, spelling, punctuation, sentence structure, and discipline-appropriate language. *5 points*	No submission or unacceptable submission *0 points*

Westcliff University Library & Digital Resources Center

As a student of Westcliff University, you gain access to a virtual library. The Westcliff University Library & Digital Resources Center provides access to an online database collection of more than 80 million scholarly journals, articles, periodicals, books, encyclopedias, newspapers, magazines, and audio and video clips. Westcliff University students are encouraged to utilize the resources in the Library & Digital Resources Center in their research for University courses.

Your course may include links to articles and videos that require you to access them through the Westcliff Library & Digital Resources Center collection. To access the Westcliff Library & Digital Resources Center collection, log in to your GAP account and click on the Library & Digital Resources Center logo on the right-hand side. For library-related questions, you may email the Library & Digital Resources Center Coordinator at library@westcliff.edu.

Accessibility and ADA Compliance

In alignment with the university's commitment to inclusive education and in accordance with international accessibility standards, this course has been designed Utilizing Web Content Accessibility Guidelines (WCAG).

Digital course materials including documents, multimedia, learning activities, and assessments are developed with accessibility in mind to ensure equitable access for all students, including those who use assistive technologies such as screen readers, keyboard navigation, or captioning services. All links will open in a new browser to ensure connectivity with the course is maintained.

Students who encounter any accessibility barriers or have difficulty accessing course content are encouraged to contact the instructor promptly. Additionally, students with documented accommodation should connect with the university's ADA Compliance Coordinator to ensure proper support is in place.

The university is committed to creating a learning environment that respects and responds to the diverse needs of its student body, and accessibility is a foundational part of this commitment.

Copyright Notice of Course Materials

Lectures and course materials, including PowerPoint presentations, tests, outlines, and similar materials, are protected by copyright. Westcliff University and individual faculty members are the exclusive owners of copyrights in those materials created. You may take notes and make copies of course materials for your own use. You may not and may not allow others to reproduce or distribute lecture notes and course materials publicly whether a fee is charged or not without express written consent of Westcliff University.

Sharing copyrighted course materials on a commercial website may violate Westcliff's Code of Conduct, where all members of the University community are expected to "…uphold the highest standards of academic integrity" and could lead to an allegation of an academic offense.

Course Learning Materials Reference List

Week 1

Analytics Vidhya. (2015). *Ultimate guide for data exploration in Python using NumPy, Matplotlib and Pandas*. https://www.analyticsvidhya.com/blog/2015/04/comprehensive-guide-data-exploration-sas-using-python-numpy-scipy-matplotlib-pandas/

Data-driven decision making: How to go from data to decisions! (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=EV3T1RYWC9g

Data-driven decision making (2025): Latest stats, trends & benchmarks. (2025). Hydrogen BI. https://www.hydrogenbi.com/data-driven-decision-making-2025-stats

Descriptive, predictive, & prescriptive analytics. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=lxCaYg1G9fE

Google Data Analytics Certificate. (n.d.). *Making data-driven decisions in business* [Video]. YouTube. https://www.youtube.com/watch?v=EevYFzAqJ2M

Gordon, M. E. (2023). *Business analytics*. SAGE Publications.

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try

Qlik. (n.d.). *What is descriptive analytics?* https://www.qlik.com/us/reporting-analytics/descriptive-analytics

Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

What are descriptive analytics? (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=DIFqQy10aCs

What is data-driven decision making? 6 key steps, importance, tools & techniques. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=-4iIC5xsiGM

Week 2

End-to-end Tableau for beginners. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=OmNeLFmyMJQ
Gordon, M. E. (2023). *Business analytics*. SAGE Publications.
Learn Tableau in 15 minutes and create your first report (FREE sample files). (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=tNLp_JVipoQ
Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students
Tableau. (n.d.). *What is data visualization? Definition, examples, and learning resources*. https://www.tableau.com/visualization/what-is-data-visualization

Week 3

Kapiche. (n.d.). *Text mining made easy: Definitions, step-by-step tutorial*. https://www.kapiche.com/blog/text-mining
Label Your Data. (n.d.). *Unstructured text in data mining: Unlocking valuable insights in document processing*. https://labelyourdata.com/articles/unstructured-text-in-data-mining
Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students
What is text mining? (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=BeDeHntF68M&t=45s
What is text mining? Explained in 10 minutes. AI-machine learning tutorial. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=V0gxfAWwJSM

Week 4

Forecasting in Excel: MUST-KNOW for any analyst. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=6rqhMnOgQnU
Gordon, M. E. (2023). *Business analytics*. SAGE Publications.
Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students
TransImpact. (n.d.). *Forecasting methods: The complete guide*. https://transimpact.com/blog/forecasting-methods/
What is time series analysis? (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=GE3JOFwTWVM

Week 5

Data Science Wednesday. (n.d.). *The fundamentals of predictive analytics* [Video]. YouTube. https://www.youtube.com/watch?v=4y6fUC56KPw
Google for Developers. (n.d.). *Logistic regression*. https://developers.google.com/machine-learning/crash-course/logistic-regression
Gordon, M. E. (2023). *Business analytics*. SAGE Publications.
Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students

Week 6

Excel forecasting: Data analysis & predictive modeling tutorial. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=D54hg-_BtuQ
Intro to business intelligence | Part 1. (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=dn97ux9exbY
Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students
Wall Street Prep. (n.d.). *Financial modeling guide | Excel training tutorial*. https://www.wallstreetprep.com/knowledge/financial-modeling/

Workday. (n.d.). *Financial forecasting in Excel made easy*. https://blog.workday.com/en-us/financial-forecasting-in-excel-made-easy.html

Week 7

Implementing a business intelligence strategy in your organization (15 minutes). (n.d.). [Video]. YouTube. https://www.youtube.com/watch?v=m1zTr7l8cwU
Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Developing a business intelligence strategy*. https://www.tableau.com/business-intelligence/business-intelligence-strategy
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students
ThoughtSpot. (n.d.). *How to develop a successful business intelligence strategy*. https://www.thoughtspot.com/data-trends/business-intelligence/business-intelligence-strategy

Week 8

Week 8 requires students to review previously assigned course materials in preparation for the Socratic Metric Reflection and the Final Comprehensive Assignment (Capstone). No new required readings are identified for this unit.

Project Jupyter. (n.d.). *Try Jupyter*. Retrieved August 14, 2026, from https://jupyter.org/try
Tableau. (n.d.). *Tableau for students*. https://www.tableau.com/academic/students